Affirmed Networks Troubleshooting Guide

Release 9.4.0.0
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- Configuring Interfaces
- Technical Support

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Preface

This guide provides information to help identify and troubleshoot common issues associated with Affirmed Networks software.

Note: *The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation staff, and other qualified service personnel responsible for installing, provisioning, and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G or 4G networks.

How to Use This Guide

This guide contains the following chapters:

Chapter	Describes
Chapter 1, Diagnosing Issues and Performing Health Checks	Procedures to check the system health and diagnose issues using the CLI.
Chapter 2, Session Management and Debugging	Troubleshooting problems related to bearer activation and bearer traffic, and recovering from a fatal error.
Chapter 3, Configuring Interfaces	Information on configuring interfaces.
Chapter 4, Technical Support	Accessing the Affirmed Networks support ticket system and escalation contacts for the Technical Assistance Center (TAC).
Appendix A, Rename corefiles to fault	Examples of corefiles to fault renaming change.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	**Times New Roman bold font**	Use **configure_ha** to configure... Enter **y**. To continue, press **Enter**.
Selecting a menu item	**Menu** => **Option**	Select **Security** => **Group Management**.
Variable parameters for which you supply values	***<courier bold italic blue font>***	**configure_dr convert slave --masterhost *<host IP>***
User input	**Courier bold blue font**	**cd /opt/Affirmed/NMS/bin**
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	**[ ]**	**cluster [id] node [id] reboot**
Optional keywords and arguments in curly brackets	**{ }**	**{*cdr-file-name*}**
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	|	Is this correct? [y|n]
Keywords representing a single element in angle brackets	**< >**	***<host name>***
Asterisk after a CLI object indicates a required object.	*****	**name***
Variables, book and chapter titles, file path names, new terms, and emphasized text	*Times New Roman italic font*	See the *Acuitas User's Guide*.

Note: *Additional information that may apply to the subject text.*

Tip: Helpful suggestions.

CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

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To contact the TAC:

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Local: 1-978-268-0871

Toll free: 1-888-283-2838

Chapter 1

1 Diagnosing Issues and Performing Health Checks

This section describes the procedures to check the system health and diagnose issues using the Command Line Interface (CLI).

1.1 Displaying Command Line Interface Command Output

CLI command output displays either horizontally or vertically, depending on the width of the terminal window. Both orientations are used in this guide. For example:

Horizontal output:

```
an3000-mcm-slot7cpu1# show extended-storage status
extended-storage status
                   ADMIN  OPER                            CURRENT    CURRENT
STATE    ID   SLOT STATE  STATUS DISK PARTITION    OPERATION  STATE
---------------------------------------------------------------------
active   125  7    unused down   extended-storage  none       none
standby  125  8    unused down   extended-storage  none       none
```

Vertical output:

```
extended-storage status active 125 7
 admin-state       unused
 oper-status       down
 disk-partition    extended-storage
 current-operation none
 current-state     none
extended-storage status standby 125 8
 admin-state       unused
 oper-status       down
 disk-partition    extended-storage
 current-operation none
 current-state     none
```

You can copy all CLI command input and output and save it to an external file. This information may be used by the Affirmed Networks Technical Assistance Center (TAC) to diagnose issues.

1.2 Subscriber Commands

Use the subscriber commands to display real-time information about subscriber data.

1.2.1 show subscriber summary

Use the **`show subscriber summary`** command to display information about individual subscribers by IMSI, APN, and other criteria.

Note: *Do not run the* **`show subscriber summary`** *command without including additional parameters. This command displays* ***all*** *subscribers and is resource intensive.*

Use the following parameters to display more information about attached subscribers:

Table 1-1 CLI Subscriber Command Parameters

Parameter	Description
active-only	Only select active subscribers
apn	Specific APN value (case insensitive)
bearer-type	Only select subscribers with a specific bearer/PDP type
control-plane-session-only	Only select control plane sessions
credited-only	Only select credited subscribers
cups-peer	Specific CUPS(Control or User Plane) Peer's name
cups-profile-id	CUPS(Control or User Plane) Profile Id
cups-session-id	Specific CUPS(Control or User Plane) Session Id Value
database-type	Only select subscribers with a specific database type
emergency-only	Only select emergency subscribers
end-imsi	End IMSI value, inclusive
exclude-indirect-forwarding-tunnels	Exclude indirect forwarding tunnels
fsapi	Only select subscribers with a specific FSAPI value
gateway-name	Gateway name
gateway-type	Gateway type

Table 1-1 CLI Subscriber Command Parameters

Parameter	Description
home-only	Only select home subscribers
idle-only	Only select idle subscribers
imsi	Specific IMSI value
indirect-forwarding-tunnels-only	Only select indirect forwarding tunnels
ip-address	Only select subscribers allocated a specific IP Address
ip-address-allocated	Only select subscribers who have IP address allocated
mac-address	MAC address
msisdn	Specific MSISDN value
rat-type	Only select subscribers using specific Radio Access Technology
start-imsi	Start IMSI value, inclusive
subscriber-nai	Subscriber NAI
uptime-greater-than	Only select subscribers up for longer than specified number of seconds
uptime-less-than	Only select subscribers up for less than specified number of seconds
using-default-quota	Only select subscriber using default quota
visiting-only	Only select visiting subscribers

For example:

```
an3000-mcm-slot7cpu1# show subscriber summary imsi 226041000000003
subscriber show-summary 98304
 imsi                       226041000000003
 gateway-name               UPLANE-1
 gateway-type               saegw-u
 meid                       999900000000030
 msisdn                     40010101700002
 apn                        sfr.consumer
 service-name               GwAgent_6_9_1_2
 subscriber-activity-state n/a
 subscriber-location-state n/a
 subscriber-nai             ""
 uptime-in-seconds          13464
 mac-address                ""
```

```
 cups-session-id            294912
 cups-peer                  SXB-V4-PEER1
 sub-memory-storage-state   in-memory
 default-bearer-id          5

an3000-mcm-slot7cpu1# show subscriber summary rat-type eutran
 imsi                       226041000000003
 gateway-name               NYsgw-1
 gateway-type               sgw-c
 meid                       999900000000030
 msisdn                     40010101700002
 apn                        sfr.consumer
 service-name               SGW_6_9_1_7
 subscriber-activity-state active
 subscriber-location-state visiting
 subscriber-nai             ""
 uptime-in-seconds          13353
 mac-address                ""
 cups-session-id            98304
 cups-peer                  CL6-UPS-V4-PEER
 sub-memory-storage-state   in-memory
 default-bearer-id          5
```

1.2.2 show subscriber pdn-session

Use the **show subscriber pdn-session** command to display information about a specific SGW or PGW session. This command is useful for troubleshooting data flows, because the PGW is the IP anchor point.

The **pdn-session** parameter displays a breakdown of a subscriber's session and provides important troubleshooting information

Note: *To view the **subscriber-type** field in CUPS deployments running MCC release 9.3 later, you must issue this command on the C-Plane. This field was removed from the U-Plane output.*

Some of the more commonly used fields for troubleshooting a subscriber's PDN session are highlighted in the following example:

```
an3000-mcm-slot8cpu1# show subscriber pdn-session 262144
subscriber pdn-session 262144
 default-bearer-id                                    5
 default-logical-bearer-id                            0
 ue-v4-ip-address                                     22.6.0.10
```

```
 peer-interface-type                                      s5
 number-of-bearers                                        1
 apn-name                                                 sfr2.consumer
 original-apn-name                                        sfr2.consumer
 prior-apn-name                                           sfr2.consumer
 incoming-apn-name                                        sfr2.consumer
 reporting-apn-name                                       sfr2.consumer
 access-in-control-teid-interface-type                    s5s8-pgw-gtpc
 access-in-control-teid-network-ctxt                      2
 access-in-control-teid-ip-address-type                   ipv4
 access-in-control-teid-v4-ip-address                     10.64.130.6
 access-in-control-teid-value                             262144
 access-out-control-teid-interface-type                   s5s8-sgw-gtpc
 access-out-control-teid-ip-address-type                  ipv4v6
 access-out-control-teid-v4-ip-address                    10.64.128.6
 access-out-control-teid-v6-ip-address
2001:470:8865::4080:6
 access-out-control-teid-value                            208897
 imsi                                                     226041000000020
 gateway-name                                             NYpgw-1
 gateway-type                                             pgw
 meid                                                     999900000000320
 msisdn                                                   999900000000320
 ULI-CGI                                                  ""
 ULI-SAI                                                  ""
 ULI-RAI                                                  ""
 ULI-TAI                                                  "MCC:208 MNC:10
TAC:291"
 ULI-ECGI                                                 "MCC:208 MNC:10
ECI:74565"
 ULI-LAI                                                  ""
 UCI-CSG-ID                                               52495
 UCI-CSG-Access-Mode                                      0
 UCI-CSG-CMI                                              1
 serving-network                                          "MCC:208 MNC:10"
bearers: { pdn-session-id bearer-id }
    262144 5
 is-cups-session                                          true
 is-pfcp-std-mode                                         false
service-data-flow-ids: { pdn-session-id service-type service-flow-id
service-rule-id }
    262144 dns-proxy 2 12
    262144 qos 2 42
nat info: { nat ip[allocation-policy] port-chunk-list }
```

```
active-connections                              6
total-connections                               20959
rejected-connections                            0
workflow-data-profile                           DATAPROF-1
workflow-control-profile                        CTRLPROF-1
workflow-override-profile                       "not set"
workflow-location-index                         5
workflow-active-chassis                         6
workflow-active-slot                            2
workflow-active-cpu                             1
workflow-gn-next-hop-network-context            2
workflow-gn-next-hop-ip                         10.64.32.1
workflow-gn-next-hop-interface-index            1
uplink-ambr                                     1316134822
downlink-ambr                                   1316134822
uplink-burst-size                               164516853
downlink-burst-size                             164516853
inactivity-timer-duration                       900
idle-timer-state                                disabled
maximal-duration-timer-state                    disabled
subscriber-type                                 4g
activation-time                          2019-06-20T12:34:03-0400
pdn-session-duration               "0 days 2 hrs 24 mins 45 secs"
pdn-session-monitor-timer-state                 running
received-pfcp-message-in-monitor-interval       true
pdn-session-state                               active
uplink-green-packets                            294366
uplink-green-bytes                              37425269
uplink-yellow-packets                           56191
uplink-yellow-bytes                             9162609
uplink-red-packets                              0
uplink-red-bytes                                0
uplink-tcp-retransmission-packets               1690
uplink-tcp-retransmission-bytes                 1317654
uplink-diverted-packets                         3712
uplink-error-drop-packets                       403
uplink-error-drop-bytes                         715529
uplink-service-drop-packets                     0
uplink-service-drop-bytes                       0
uplink-no-tft-packets                           0
uplink-no-route-packets                         0
uplink-ttl-exhaust-packets                      0
uplink-bad-header-packets                       0
```

```
uplink-bad-ip-addr-packets                      0
uplink-mtu-discard-packets                      403
uplink-transmit-icmp-packets                    0
uplink-send-error-packets                       0
uplink-send-error-bytes                         0
uplink-router-solicitation-packets              0
Uplink-router-solicitation-bytes                0
uplink-neighbour-solicitation-packets           0
uplink-neighbour-solicitation-bytes             0
downlink-green-packets                          321805
downlink-green-bytes                            201343240
downlink-yellow-packets                         72107
downlink-yellow-bytes                           81526465
downlink-red-packets                            0
downlink-red-bytes                              0
downlink-tcp-retransmission-packets             4093
downlink-tcp-retransmission-bytes               4589291
downlink-diverted-returned-packets              0
downlink-error-drop-packets                     0
downlink-error-drop-bytes                       0
downlink-service-drop-packets                   17
downlink-service-drop-bytes                     1200
downlink-no-tft-packets                         0
downlink-no-route-packets                       0
downlink-ttl-exhaust-packets                    0
downlink-mtu-discard-packets                    0
downlink-transmit-icmp-packets                  0
downlink-bad-header-packets                     0
downlink-send-error-packets                     0
downlink-send-error-bytes                       0
downlink-router-advertisement-packets           0
downlink-router-advertisement-bytes             0
downlink-neighbour-advertisement-packets        0
downlink-neighbour-advertisement-bytes          0
uplink-hairpin-policy-discard-packets           0
uplink-hairpin-error-discard-packets            0
uplink-tunneled-packets                         0
uplink-aggregate-bw-pkts-dropped                0
uplink-aggregate-bw-bytes-dropped               0
downlink-aggregate-bw-pkts-dropped              0
downlink-aggregate-bw-bytes-dropped             0
uplink-appgrp-aggregate-bw-pkts-dropped         0
uplink-appgrp-aggregate-bw-bytes-dropped        0
```

```
downlink-appgrp-aggregate-bw-pkts-dropped        0
downlink-appgrp-aggregate-bw-bytes-dropped       0
uplink-flowauth-pkts-dropped                     0
uplink-flowauth-bytes-dropped                    0
downlink-flowauth-pkts-dropped                   0
downlink-flowauth-bytes-dropped                  0
security-did-not-forward-pkts-dropped            0
rat-type                                         E-UTRAN
pdn-type                                         ipv4
workflow-subscriber-analyzer                     ANALYZE-1
user-plane-selector-name                         ""
workflow-data-profile-sub-analyzer               ANALYZE-1
l2tp-tunnel-endpoint                             ::
l2tp-tunnel-id                                   0
l2tp-session-id                                  0
l2tp-local-tunnel-id                             0
l2tp-local-session-id                            0
l2tp-tunnel-assignment-id                        ""
delay-tolerant-pdn-connection-enabled            false
low-priority-access-indicated-pdn                false
control-plane-only-pdn-connection                false
suspended                                        false
guard-timer-running                              false
time-zone                                 "GMT +1:00 hours (0x40)"
emergency-session                                false
user-identifier-type                             imsi
tethering                                        false
origination-timestamp                            0
max-wait-time                                    0
access-point-mac-address                         ""
total-tethered-connections                       0
active-tethered-connections                      0
pcscf-table-name                                 N/A
pco-ipv4-link-mtu-size                           0
pco-nonip-link-mtu-size                          0
pco-pcscf-fqdn-used-for-dns-query                ""
uplink-splmn-pkt-rate-valid                      false
downlink-splmn-pkt-rate-valid                    false
apn-pkt-rate-timeunit                            not-set
apn-pkt-rate-exception-report                    not-set
uplink-apn-pkt-rate-valid                        false
cc-group                                         ""
pco-action                                       0
```

```
is-using-default-quota                                  false
crbn-profiles                                           ""
charging-rule-names                                     " "
aggregate-qos-policy-id                                 0
uplink-peak-throughput                                  0
downlink-peak-throughput                                0
uplink-activity-duration                                0
downlink-activity-duration                              0
session-terminate-pkts-dropped                          0
proxy-bypassed-connections                              0
session-terminate-bytes-dropped                         0
tethered-flows-detected-using-ttl                       0
tethered-flows-detected-using-user-agent                0
tethering-entitled                                      false
tethering-entitled-flows                                0
content-filter-instance-name                            ""
tethered-flows-detected-using-tcp-ip-fingerprinting 0
pcrf-update-down-avoid-pcrf                             false
```

1.2.3 show subscriber pdn-service-data-flow

When troubleshooting issues specifically related to QoS, proxy, and charging, use parameters that control a particular subscriber flow.

The output from show subscriber pdn-session (page 1-4) (highlighted in blue above) displays the flow details:

```
bearers: { pdn-session-id bearer-id }
    262144 5
 is-cups-session                                        true
 is-pfcp-std-mode                                       false
service-data-flow-ids: { pdn-session-id service-type service-flow-id
service-rule-id }
    262144 dns-proxy 2 12
    262144 qos 2 42
```

This information is used to determine whether the user is, for example, using the correct service-flow-name. This command also displays specific information about workflow and service data types. The subscriber trace does not include public post-NAT UE IP address information.

Note: *This command only displays the current status of the subscriber information, which is from the SGW/PGW side pre-Proxy. No NAT mapping information is available in the current trace feature.*

Use the **show subscriber pdn-service-data-flow** command and parameters to display specific individual service data flows. For example:

```
an3000-mcm-slot7cpu1# show subscriber pdn-service-data-flow 262144
Description: PDN Service Data Flow Level Information (key: <pdn-
session-id> <service-type> <service-flow-id> <service-rule-id>)
Possible completions:
 accounting         charging           content-cache
 content-filter     dhcp               dns-proxy
 latency            metering           mirror
 nat                proxy              qos
 sip-proxy          smtp-proxy         steering
 unknown            video-adaptation   wap-gateway
 |                  <cr>
```

```
an3000-mcm-slot8cpu1# show subscriber pdn-service-data-flow 262144 qos
2 42
subscriber pdn-service-data-flow 262144 qos 2 42
 service-flow-name     QOS-FLOW-PARG
 service-rule-name     SR-PARG
 imsi                  226041000000020
 gateway-name          NYpgw-1
 gateway-type          pgw
 meid                  999900000000320
 msisdn                999900000000320
 correlation-id        0
 service-data-flow-id  35
 total-connections     3512
 total-packets         128550
 total-drop-packets    0
```

1.2.4 show subscriber count

Use the **show subscriber count** command to display the global number of subscribers.

Use the parameters in Table 1-1 to define a specific category or range of subscribers.

For example:

```
an3000-mcm-slot7cpu1# show subscriber count <category/range>
```

1.2.5 subscriber clear

In certain situations, you may need to clear user(s) from the MCC. Use the **subscriber clear** command and the parameters in Table 1-1.

For example:

```
an3000-mcm-slot7cpu1# subscriber clear imsi 123456789876543
```

1.3 Cluster Commands

Use the cluster commands to diagnose issues and display statistics.

The primary OAM interface for each cluster is through the management node (the MCM), which is implemented on nodes (VMs) 7 and 8 on each cluster.

1.3.1 show cluster summary

Use the **show cluster summary** command to display information about each VM, including admin state, HA state, and uptime.

An asterisk at the end of the release number (highlighted in green below) indicates that one or more tasks on the node is not yet operationally up. Depending on the configuration, some tasks can take longer than others to come up. When all nodes are operationally up, the cluster is fully operational.

A CPU state of down (highlighted in yellow below) indicates a more general fault on the VM or blade which could mean there is a problem with the underlying hypervisor layer (VMware) or a physical issue on the blade.

The asterisks are also seen during a VM reboot as processes startup, but are mostly seen during genuine fault scenarios (or conditions).

Note: *If all the VMs on one physical blade are down, the problem is likely to be on the hypervisor layer or the physical blade.*

```
an3000-mcm-slot7cpu1# show cluster summary
   SLOT   CPU                                    ADMIN               CPU
ID NUMBER NUMBER TYPE   PERSONALITY MODEL  STATE     HA STATE   STATE  CPU UPTIME             RELEASE
-------------------------------------------------------------------------------------------------------------
71 1      1      v-csm  CSM         vCSM   enabled   active     up     D:007 H:14 M:46 S:30   9.3.0.0-678.REL
   5      1      v-ssm  SSM         vSSM   enabled   active     up     D:007 H:14 M:46 S:24   9.3.0.0-678.REL*
   7      1      v-mcm  MCM         vMCM   enabled   primary    up     D:007 H:14 M:47 S:06   9.3.0.0-678.REL
   8      1      v-mcm  MCM         vMCM   enabled   secondary  down   D:000 H:00 M:00 S:00
```

1.3.2 show cluster node task

Use the **show cluster node task** command to display the status of the task running under each node and its associated service. For example:

```
an3000-mcm-slot7cpu1# show cluster node task
cluster 100
 node 1
  task dryup
CPU
DRYUP                                                                        HA     OPER
NUMBER TASK NAME                             SERVICE NAME                    STATE  STATE DRYUP STATUS   LOAD
----------------------------------------------------------------------------------------------------------------
cpu1   EventConfigRelayAgent_100_1_1_
       EventConfigRelayAgent_2               EventConfigRelayAgent_100_1_1   active  up   Not applicable 0
cpu1   FMAgent_100_1_1_FMAgent_2             FMAgent_100_1_1                 active  up   Not applicable 0
cpu1   FPexecMaint_100_1_1_FPexecMaint_0     FPexecMaint_100_1_1             active  up   Not applicable 0
cpu1   FpUeIpHdlr_100_1_1_0_FpUeIpHdlr_0     FpUeIpHdlr_100_1_1_0            active  up   Not applicable 0
cpu1   FpUeIpHdlr_100_1_1_1_FpUeIpHdlr_1     FpUeIpHdlr_100_1_1_1            active  up   Not applicable 0
cpu1   GwStatsAgent_100_1_1_GwStatsAgent_2   GwStatsAgent_100_1_1            active  up   Not applicable 0
cpu1   HwAgent_100_1_1_HwAgent_2             HwAgent_100_1_1                 active  up   Not applicable 0
cpu1   ImAgent_100_1_1_ImAgent_2             ImAgent_100_1_1                 active  up   Not applicable 0
cpu1   NRA_100_1_1_NRA_2                     NRA_100_1_1                     active  up   Not applicable 0
cpu1   NSAgent_100_1_1_NSAgent_0             NSAgent_100_1_1                 active  up   Not applicable 0
cpu1   SecPolicyAgt_100_1_1_SecPolicyAgt_0   SecPolicyAgt_100_1_1            active  up   Not applicable 0
cpu1   TosfpFlowAgt_100_1_1_TosfpFlowAgt_0   TosfpFlowAgt_100_1_1            active  up   Not applicable 0
cpu1   TosfpGwHdlr_100_1_1_0_TosfpGwHdlr_0   TosfpGwHdlr_100_1_1_0           active  up   Not applicable 0
cpu1   TosfpGwHdlr_100_1_1_1_TosfpGwHdlr_1   TosfpGwHdlr_100_1_1_1           active  up   Not applicable 0
cpu1   TosfpTblDist_TosfpTblDist_0           TosfpTblDist                    active  up   Not applicable 0
cpu1   TransProxyAgt_100_1_1_TransProxyAgt_0 TransProxyAgt_100_1_1           active  up   Not applicable 0

  task operation-detail
CPU                                                                          HA     OPER
NUMBER TASK NAME                             SERVICE NAME                    STATE  STATE OPER REASON
----------------------------------------------------------------------------------------------------------
cpu1   EventConfigRelayAgent_100_1_1_
       EventConfigRelayAgent_2               EventConfigRelayAgent_100_1_1   active  up   Not applicable
cpu1   FMAgent_100_1_1_FMAgent_2             FMAgent_100_1_1                 active  up   Not applicable
cpu1   FPexecMaint_100_1_1_FPexecMaint_0     FPexecMaint_100_1_1             active  up   Not applicable
cpu1   FpUeIpHdlr_100_1_1_0_FpUeIpHdlr_0     FpUeIpHdlr_100_1_1_0            active  up   Not applicable
cpu1   FpUeIpHdlr_100_1_1_1_FpUeIpHdlr_1     FpUeIpHdlr_100_1_1_1            active  up   Not applicable
cpu1   GwStatsAgent_100_1_1_GwStatsAgent_2   GwStatsAgent_100_1_1            active  up   Not applicable
cpu1   HwAgent_100_1_1_HwAgent_2             HwAgent_100_1_1                 active  up   Not applicable
cpu1   ImAgent_100_1_1_ImAgent_2             ImAgent_100_1_1                 active  up   Not applicable
cpu1   NRA_100_1_1_NRA_2                     NRA_100_1_1                     active  up   Not applicable
cpu1   NSAgent_100_1_1_NSAgent_0             NSAgent_100_1_1                 active  up   Not applicable
cpu1   SecPolicyAgt_100_1_1_SecPolicyAgt_0   SecPolicyAgt_100_1_1            active  up   Not applicable
cpu1   TosfpFlowAgt_100_1_1_TosfpFlowAgt_0   TosfpFlowAgt_100_1_1            active  up   Not applicable
cpu1   TosfpGwHdlr_100_1_1_0_TosfpGwHdlr_0   TosfpGwHdlr_100_1_1_0           active  up   Not applicable
cpu1   TosfpGwHdlr_100_1_1_1_TosfpGwHdlr_1   TosfpGwHdlr_100_1_1_1           active  up   Not applicable
cpu1   TosfpTblDist_TosfpTblDist_0           TosfpTblDist                    active  up   Not applicable
cpu1   TransProxyAgt_100_1_1_TransProxyAgt_0  TransProxyAgt_100_1_1          active  up   Not applicable

  task statistics
CPU                                                                          CPU          MEMORY
NUMBER TASK NAME                                                             UTILIZATION  UTILIZATION
----------------------------------------------------------------------------------------------------------
cpu1   EventConfigRelayAgent_100_1_1_EventConfigRelayAgent_2                 0.05         0.07
cpu1   FMAgent_100_1_1_FMAgent_2                                             0.05         0.08
cpu1   FPexecMaint_100_1_1_FPexecMaint_0                                     9.5075       4.3525
cpu1   FpUeIpHdlr_100_1_1_0_FpUeIpHdlr_0                                     9.5075       4.3525
cpu1   FpUeIpHdlr_100_1_1_1_FpUeIpHdlr_1                                     9.5075       4.3525
cpu1   GwStatsAgent_100_1_1_GwStatsAgent_2                                   0.05         0.1
cpu1   HwAgent_100_1_1_HwAgent_2                                             0.05         0.1
cpu1   ImAgent_100_1_1_ImAgent_2                                             0.05         0.07
cpu1   NRA_100_1_1_NRA_2                                                     0.06         0.1
cpu1   NSAgent_100_1_1_NSAgent_0                                             0.06         0.09
cpu1   SecPolicyAgt_100_1_1_SecPolicyAgt_0                                   9.5075       4.3525
cpu1   TosfpFlowAgt_100_1_1_TosfpFlowAgt_0                                   9.5075       4.3525
cpu1   TosfpGwHdlr_100_1_1_0_TosfpGwHdlr_0                                   9.5075       4.3525
cpu1   TosfpGwHdlr_100_1_1_1_TosfpGwHdlr_1                                   9.5075       4.3525
cpu1   TosfpTblDist_TosfpTblDist_0                                           0.05         0.07
cpu1   TransProxyAgt_100_1_1_TransProxyAgt_0                                 9.5075       4.3525
```

1.3.3 show cluster node <node-id> task operation-detail oper-state

Use the **show cluster node *<node-id>* task** command to display processes at the node level and show details to help identify issues related to a specific process.

Use the following parameters to display the operational status of tasks:

```
an3000-mcm-slot7cpu1# show cluster node {node-id} task operation-
detail oper-state [congested|down|dryup|maint|partially-up|up]
```

For example:

```
an3000-mcm-slot7cpu1# show cluster node task operation-detail oper-
state
cluster
         CPU                                                       OPER
ID   ID  NUMBER  TASK NAME                                         STATE
------------------------------------------------------------------------------
100  1   cpu1    EventConfigRelayAgent_100_1_1_EventConfigRelayAgent_2  up
         cpu1    FMAgent_100_1_1_FMAgent_2                         up
         cpu1    FPexecMaint_100_1_1_FPexecMaint_0                 up
         cpu1    FpUeIpHdlr_100_1_1_0_FpUeIpHdlr_0                 up
         cpu1    FpUeIpHdlr_100_1_1_1_FpUeIpHdlr_1                 up
         cpu1    GwStatsAgent_100_1_1_GwStatsAgent_2               up
         cpu1    HwAgent_100_1_1_HwAgent_2                         up
         cpu1    ImAgent_100_1_1_ImAgent_2                         up
         cpu1    NRA_100_1_1_NRA_2                                 up
         cpu1    NSAgent_100_1_1_NSAgent_0                         up
         cpu1    SecPolicyAgt_100_1_1_SecPolicyAgt_0               up
         cpu1    TosfpFlowAgt_100_1_1_TosfpFlowAgt_0               up
         cpu1    TosfpGwHdlr_100_1_1_0_TosfpGwHdlr_0               up
         cpu1    TosfpGwHdlr_100_1_1_1_TosfpGwHdlr_1               up
         cpu1    TosfpTblDist_TosfpTblDist_0                       up
         cpu1    TransProxyAgt_100_1_1_TransProxyAgt_0             up
     2   cpu1    EventConfigRelayAgent_100_2_1_EventConfigRelayAgent_3  up
         cpu1    FMAgent_100_2_1_FMAgent_3                         up
         cpu1    FPexecMaint_100_2_1_FPexecMaint_1                 up
         cpu1    FpUeIpHdlr_100_2_1_2_FpUeIpHdlr_2                 up
         cpu1    FpUeIpHdlr_100_2_1_3_FpUeIpHdlr_3                 up
         cpu1    GwStatsAgent_100_2_1_GwStatsAgent_3               up
         cpu1    HwAgent_100_2_1_HwAgent_3                         up
         cpu1    ImAgent_100_2_1_ImAgent_3                         up
         cpu1    NRA_100_2_1_NRA_3                                 up
         cpu1    NSAgent_100_2_1_NSAgent_1                         up
         cpu1    SecPolicyAgt_100_2_1_SecPolicyAgt_1               up
         cpu1    TosfpFlowAgt_100_2_1_TosfpFlowAgt_1               up
         cpu1    TosfpGwHdlr_100_2_1_2_TosfpGwHdlr_2               up
         cpu1    TosfpGwHdlr_100_2_1_3_TosfpGwHdlr_3               up
         cpu1    TransProxyAgt_100_2_1_TransProxyAgt_1             up
```

1.3.4 show cluster statistics

Use the **show cluster statistics** command to display details of each VM's memory, CPU utilization and process count. Both CPU and memory should be no higher than 80% for the MCC to function predictably.

The SSM nodes dedicate a large percentage of their cores to packet handling loops to avoid certain performance penalties caused by using the Linux kernel. This command displays system status from a Linux perspective.

```
an3000-mcm-slot7cpu1# show cluster statistics
cluster
                                                                     ISOLATED
                                                                     CPU
          CPU      CPU          MEMORY       PROCESS   UTILIZATION  GENERAL CPU
ID  SLOT  NUMBER   UTILIZATION  UTILIZATION  COUNT     NORMALIZED   UTILIZATION
-------------------------------------------------------------------------------
18  1     1        28.94        36.06        277       00.00        18.78
    5     1        60.78        38.47        211       00.00        04.11
    7     1        01.65        39.85        256       00.00        01.65
```

For more details, see *Slot Statistics* in *Acuitas Performance Statistics*.

Use the **show zone default system statistics** command to display CPU utilization that has been adjusted by the system to account for the internal architecture decisions, and displays a more realistic value of the CPU utilization.

1.3.5 show cluster control-plane protection status

Use the **control-plane-protection status** command to display the control plane redundancy table of communication of the VMs over the base network. The base network is responsible for inter-VM control communication; no data traffic is transferred on this path. Each VM has two redundant paths to the base network using port-A or port-B.

```
an3000-mcm-slot7cpu1# show cluster control-plane-protection status
                                BITMAP          BITMAP
    NODE  USING   NUM           REACHABLE       REACHABLE
ID  ID    PORT    SWITCHOVERS   NODES PORT A    NODES PORT B    MOST RECENT SWITCHOVER CAUSE
--------------------------------------------------------------------------------------------------------
125 1     port-B  5             0x79df          0x79df          2019-06-20T13:51:32-0400 Switching to path:B
                                                                map: ffff, (A=eth0.402, B=eth2.403), active
                                                                MCM path B
    2     port-B  1             0x79df          0x79df          2019-06-12T13:59:45-0400 Switching to path:B
                                                                map: ffff, (A=eth0.402, B=eth2.403), active
                                                                MCM path B
    3     port-B  7             0x79df          0x79df          2019-06-15T02:47:08-0400 Switching to path:B
                                                                map: ffff, (A=eth0.402, B=eth2.403), active
                                                                MCM path B
```

Note: *The nodes on the system will be on path A (MCM-7) or path B (MCM-8) depending on which MCM is active.*

1.3.6 show cluster data-plane protection status

Use the **`data-plane-protection status`** command to display the data plane redundancy table relating to VMs on the EW (East-West traffic) network. Each VM has two redundant paths for EW communication.

```
an3000-mcm-slot7cpu1# show cluster data-plane-protection status
cluster
                                BITMAP          BITMAP
     NODE  USING    NUM         REACHABLE       REACHABLE
ID   ID    PORT     SWITCHOVERS NODES PORT A    NODES PORT B     MOST RECENT SWITCHOVER CAUSE
-----------------------------------------------------------------------------------------------------
125  1     port-B   2           0x791f          0x791f           2019-06-19T16:30:01-0400 Switching to path:B
                                                                 map: [f-f], (A=fab0.604, B=fab1.605), active
                                                                 MCM path B
     2     port-B   2           0x791f          0x791f           2019-06-19T16:30:01-0400 Switching to path:B
                                                                 map: [f-f], (A=kni0.604, B=kni1.605), active
                                                                 MCM path B
```

1.3.7 show cluster control-plane-protection keepalive

Use the **`keepalive`** parameter with the **`data-plane-protection`** and **`control-plane-protection`** commands to display a matrix of the response times between each VM (NODE ID) and its cluster neighbor (PEER NODE ID) over both redundant paths.

Round Trip Times (RTT) greater than 300ms should be investigated. Extremely high RTT, e.g. 15324, or a value of 0, usually indicates that communication on that link is broken and requires urgent attention.

```
an3000-mcm-slot7cpu1# show cluster control-plane-protection keepalive
cluster
                                  5          5                             5          5
                         PORT A   MINUTE     MINUTE              PORT B    MINUTE     MINUTE
                PORT A   MOST     HISTORY    HISTORY   PORT B    MOST      HISTORY    HISTORY
         PEER   MOST     RECENT   PERCENT    PERCENT   MOST      RECENT    PERCENT    PERCENT
    NODE NODE   RECENT   RX+TX    RX PORT    RX+TX     RECENT    RX TX     RX PORT    RX+TX
ID  ID   ID     RX MS    MS       A          PORT A    RX MS     MS        B          PORT B
---------------------------------------------------------------------------------------------
11  1    3      188      188      100        100       188       188       100        100
    1    7      250      250      100        100       250       250       100        100
    3    1      1        1        100        100       251       251       100        100
    3    7      1        1        100        100       251       251       100        100
    7    1      250      250      100        100       250       250       100        100
    7    3      188      188      100        100       188       188       100        100
```

1.3.8 show cluster data-plane-protection keepalive

```
an3000-mcm-slot7cpu1# show cluster data-plane-protection keepalive
cluster
                                  5          5                             5          5
                         PORT A   MINUTE     MINUTE              PORT B    MINUTE     MINUTE
                PORT A   MOST     HISTORY    HISTORY   PORT B    MOST      HISTORY    HISTORY
         PEER   MOST     RECENT   PERCENT    PERCENT   MOST      RECENT    PERCENT    PERCENT
    NODE NODE   RECENT   RX+TX    RX PORT    RX+TX     RECENT    RX+TX     RX PORT    RX+TX
ID  ID   ID     RX MS    MS       A          PORT A    RX MS     MS        B          PORT B
----------------------------------------------------------------------------------------------
11  1    3      1        1        100        100       1         1         100        100
    3    1      14       14       100        100       14        14        100        100
```

1.3.9 show cluster <id> node <node-id> task statistics

Use this command to view the memory and CPU utilization on a node level within a cluster. This information is useful in isolating instances where resources are oversubscribed at a task level.

```
an3000-mcm-slot7cpu1# show cluster 11 node 3 task statistics memory-
utilization
task statistics
CPU                                                        MEMORY
NUMBER   TASK NAME                                         UTILIZATION
-----------------------------------------------------------------------------
cpu1     EventConfigRelayAgent_11_3_1_EventConfigRelayAgent_3   0.12
cpu1     FMAgent_11_3_1_FMAgent_3                          0.12
cpu1     FPexecMaint_11_3_1_FPexecMaint_0                  1.2275
cpu1     FileAgent_11_3_1_FileAgent_3                      0.13
cpu1     FpUeIpHdlr_11_3_1_0_FpUeIpHdlr_0                  1.2275
cpu1     FpUeIpHdlr_11_3_1_1_FpUeIpHdlr_1                  1.2275
cpu1     GwStatsAgent_11_3_1_GwStatsAgent_3                0.16
cpu1     HwAgent_11_3_1_HwAgent_3                          0.16
cpu1     ImAgent_11_3_1_ImAgent_3                          0.12
cpu1     NRA_11_3_1_NRA_3                                  0.17
cpu1     NSAgent_11_3_1_NSAgent_1                          0.13
cpu1     NmdAgent_11_3_1_NmdAgent_3                        0.11
cpu1     SecPolicyAgt_11_3_1_SecPolicyAgt_0                1.2275
cpu1     TosfpFlowAgt_11_3_1_TosfpFlowAgt_0                1.2275
cpu1     TosfpGwHdlr_11_3_1_0_TosfpGwHdlr_0                1.2275
cpu1     TosfpGwHdlr_11_3_1_1_TosfpGwHdlr_1                1.2275
cpu1     TosfpTblDist_TosfpTblDist_0                       0.12
cpu1     TosfpTblDist_TosfpTblDist_1                       0.11
cpu1     TransProxyAgt_11_3_1_TransProxyAgt_0              1.2275
cpu1     UrlListMgr_11_3_1_UrlListMgr_0                    0.4
```

```
an3000-mcm-slot7cpu1# show cluster 100 node 21 task statistics cpu-
utilization
task statistics
CPU                                                        CPU
NUMBER   TASK NAME                                         UTILIZATION
----------------------------------------------------------------------------
cpu1     EventConfigRelayAgent_11_3_1_EventConfigRelayAgent_3   0.05
cpu1     FMAgent_11_3_1_FMAgent_3                          0.05
cpu1     FPexecMaint_11_3_1_FPexecMaint_0                  7.975
cpu1     FileAgent_11_3_1_FileAgent_3                      0.06
cpu1     FpUeIpHdlr_11_3_1_0_FpUeIpHdlr_0                  7.975
cpu1     FpUeIpHdlr_11_3_1_1_FpUeIpHdlr_1                  7.975
cpu1     GwStatsAgent_11_3_1_GwStatsAgent_3                0.05
cpu1     HwAgent_11_3_1_HwAgent_3                          0.05
cpu1     ImAgent_11_3_1_ImAgent_3                          0.05
cpu1     NRA_11_3_1_NRA_3                                  0.05
```

```
cpu1     NSAgent_11_3_1_NSAgent_1                                0.05
cpu1     NmdAgent_11_3_1_NmdAgent_3                              0.05
cpu1     SecPolicyAgt_11_3_1_SecPolicyAgt_0                      7.975
cpu1     TosfpFlowAgt_11_3_1_TosfpFlowAgt_0                      7.975
cpu1     TosfpGwHdlr_11_3_1_0_TosfpGwHdlr_0                      7.975
cpu1     TosfpGwHdlr_11_3_1_1_TosfpGwHdlr_1                      7.975
cpu1     TosfpTblDist_TosfpTblDist_0                             0.05
cpu1     TosfpTblDist_TosfpTblDist_1                             0.05
cpu1     TransProxyAgt_11_3_1_TransProxyAgt_0                    7.975
cpu1     UrlListMgr_11_3_1_UrlListMgr_0                          0.05
```

1.3.10 show cluster node fault-summary

When a software component of the MCC faults it generates a fault file, which is essentially a memory dump at the time of the fault. This file helps the TAC to identify and resolve the root cause of the issue. The fault file is stored in */public/corefiles/* which is accessible through the MCM Linux shell.

```
an3000-mcm-slot7cpu1# show cluster node fault-summary
      SECONDS
      SINCE
ID ID EPOCH       FAULT FILENAME                                   TIME
---------------------------------------------------------------------------------------------------
63 3  1560879981  slot3cpu1/fault_6986_1560879972_PDNGW_63_3_1_2_.gz   Tue 18 Jun 2019 01:46:21 PM EDT
      1560879981  slot3cpu1/fault_7018_1560879971_PDNGW_63_3_1_5_.gz   Tue 18 Jun 2019 01:46:21 PM EDT
      1560879982  slot3cpu1/fault_6927_1560879972_PDNGW_63_3_1_3_.gz   Tue 18 Jun 2019 01:46:22 PM EDT
      1560879982  slot3cpu1/fault_6947_1560879972_PDNGW_63_3_1_0_.gz   Tue 18 Jun 2019 01:46:22 PM EDT
      1560879982  slot3cpu1/fault_6965_1560879972_PDNGW_63_3_1_7_.gz   Tue 18 Jun 2019 01:46:22 PM EDT
      1560879982  slot3cpu1/fault_6982_1560879972_PDNGW_63_3_1_1_.gz   Tue 18 Jun 2019 01:46:22 PM EDT
```

1.3.11 show cluster node disk

Use the **disk** parameter to display the disk size, utilization, and error status of the VMs within the MCC cluster.

```
an3000-mcm-slot7cpu1# show cluster node disk
cluster
                                 ADMIN       OPER  DISK      DISK     USE      DISK
ID   ID   DISKID  NAME           STATE       STATUSSIZE      USAGE    PERCENT  ERROR
---------------------------------------------------------------------------------------
11   1    1       proxy-cache        enabled     up    -         -        -        No
          2       storage            enabled     up    9.2G      150M     2%       No
          3       images             enabled     up    7.4G      145M     3%       No
          4       rollback-image     enabled     up    9.2G      149M     2%       No
          5       prepared-image     enabled     up    9.2G      4.2G     48%      No
          6       operational-image  enabled     up    9.2G      149M     2%       No
          8       ram-storage        disabled    down  -         -        -        No
     3    2       storage            enabled     up    9.2G      150M     2%       No
          3       images             enabled     up    97723394  -        -        No
          4       rollback-image     enabled     up    9.2G      149M     2%       No
```

```
        5     prepared-image      enabled    up    9.2G    4.2G    48%    No
        6     operational-image   enabled    up    9.2G    149M    2%     No
        8     ram-storage         disabled   down  -       -       -      No
    7   2     storage             enabled    up    28G     403M    2%     No
        3     images              enabled    up    19G     172M    1%     No
        4     rollback-image      enabled    up    19G     172M    1%     No
        5     prepared-image      enabled    up    19G     12G     67%    No
        6     operational-image   enabled    up    19G     172M    1%     No
        7     extended-storage    unknown    down  -       -       -      -
        8     ram-storage         disabled   down  -       -       -      No
```

1.3.12 cluster <id> clear-faults

After collecting the fault files, use the **clear-faults** parameter to clear the files from the system.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 clear-faults
```

Normally, you will also need to clear the associated alarm.

For example:

```
an3000-mcm-slot7cpu1# alarm active 223232 clear
```

1.3.13 cluster <id> reboot

Use the **cluster** ***<id>*** **reboot** command to reboot the entire virtual chassis. Issue this command with caution to avoid data loss.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 reboot
```

1.3.14 cluster <id> node <node-id> reboot

Use the **cluster** ***<id>*** **node** ***<node-id>*** **reboot** command to reboot a specific node (VM) on the cluster.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 node 3 reboot
```

1.3.15 cluster <id> node <node-id> processes

Use the parameters in Table 1-2 to perform the relevant action on the node processes.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 node 2 processes cpu1
HttpProxy_100_42_1_138_HttpProxy_138 restart
```

Table 1-2 Cluster Node Processes Parameters

Parameter	Description
clearrestartdebug	Restart this application in normal mode after debugging
fault	Kill a task and dump its fault file
force-restart	Force restart this application
maintenance	Cause this process to stop accepting new work and shed existing work
ping	Ping a task
restart	Restart this application
restartdebug	Restart this application in debug mode

1.3.16 cluster <id> mcmswitchover

Maintenance activities or issues may require an MCM switchover, making the standby MCM active. Use the **cluster *<id>* mcmswitchover** command to perform a graceful handover between the two nodes.

For example:

```
an3000-mcm-slot7cpu1# cluster 12 mcmswitchover
```

1.4 System Commands

Use the system commands to display information related to the system.

1.4.1 show image-management release-modifier-status

Use the **show image-management release-modifier-status** to display the DPI signature filename.

```
an3000-mcm-slot8cpu1# show image-management release-modifier-status

image-management release-modifier-status
RELEASE                          RELEASE    OPERATIONAL          ROLLBACK
ROLE         RELEASE VERSION     MODIFIER   VERSION              VERSION         ORIGINAL VERSION
--------------------------------------------------------------------------------------------------
rollback     9.2.0.0-865.REL     dpisig     dpisig_2019-02-22    NotAvailable    dpisig_2019-02-22
rollback     9.2.0.0-865.REL     maint      maint_2017-06-12     NotAvailable    maint_2017-06-12
rollback     9.2.0.0-865.REL     plugin     plugin_2019-03-06    NotAvailable    plugin_2019-03-06
operational  9.3.0.0-812.REL     dpisig     dpisig_2019-05-24    NotAvailable    dpisig_2019-05-24
operational  9.3.0.0-812.REL     maint      maint_2017-06-12     NotAvailable    maint_2017-06-12
operational  9.3.0.0-812.REL     plugin     plugin_2019-06-10    NotAvailable    plugin_2019-06-10
```

1.4.2 show infrastructure ntp-server status

Use the **show infrastructure ntp-server** command to display the health statistics of the connection between the MCC and the configured NTP servers.

```
an3000-mcm-slot7cpu1# show infrastructure ntp-server status
infrastructure ntp-server status
REMOTE SYNC REFID   ST  T  WHEN  POLL  REACH  DELAY  OFFSET  JITTER
----------------------------------------------------------------------
master      .INIT.  16  u  0     1024  0      0.000  0.000   0.000
```

1.4.3 show infrastructure file-management statistics

Use the **show infrastructure file-management statistics** command to display the file management activities on the MCC. Pay careful attention to the FILES PURGED DISK FULL column because an incremented value indicates a full file system and possible irretrievable data loss.

```
an3000-mcm-slot7cpu1# show infrastructure file-management statistics
infrastructure file-management statistics
                                                             FILES
                                                     FILES   PURGED
                                        FILES        PURGED  DISK
PREFIX                                  TRANSFERRED  AGING   FULL
--------------------------------------------------------------------
//boot/partitions/storage/*             0            0       0
//boot/partitions/storage/dump/*        0            0       0
//var/log/asp/*                         0            126454  0
//var/log/psm/*                         0            0       0
/var/corefiles/*                        0            0       0
/var/log/datarecord.lnk/HTDR_           0            0       0
/var/log/eventlog.lnk/*                 0            0       0
/var/log/eventlog.lnk/IMSITRACE_PCAPS   0            0       0
/var/log/eventlog.lnk/critical          0            0       0
```

1.4.4 show zone default system statistics

Use the **show zone default system statistics** command to display an extensive system level overview of the statistics of the VMs within the cluster. Use the **core-level**, **cpu-level**, and **node-level** parameters to display a more manageable subset of the required output.

For example:

```
an3000-mcm-slot7cpu1# show zone default system statistics
system statistics node-level cpu-stats

             AVG   AVG   AVG    AVG   AVG   AVG    AVG   AVG   AVG
             1MIN  5MIN  15MIN  1MIN  5MIN  15MIN  1MIN  5MIN  15MIN
             CSM   CSM   CSM    SSM   SSM   SSM    MCM   MCM   MCM
```

```
              CPU  CPU  CPU   CPU  CPU  CPU   CPU  CPU  CPU
CHASSIS SLOT UTIL UTIL UTIL  UTIL UTIL UTIL  UTIL UTIL UTIL
------------------------------------------------------------------
11      1    5    5    5     0    0    0     0    0    0
11      3    0    0    0     0    0    0     0    0    0
11      7    0    0    0     0    0    0     2    2    2
```

```
system statistics node-level memory-stats 11 1
Average 1 minute csm memory utilization                          43
Average 5 minute csm memory utilization                          43
Average 15 minute csm memory utilization                         43
Average 1 minute ssm memory utilization                           0
Average 5 minute ssm memory utilization                           0
Average 15 minute ssm memory utilization                          0
Average 1 minute mcm memory utilization                           0
Average 5 minute mcm memory utilization                           0
Average 15 minute mcm memory utilization                          0
Total available memory                                     32814880
Total available memory free                                18448240
Total available memory in buffer cache                       143548
Total available memory in page cache minus swap cache       3577692
Total available memory in swap cache                              0
Total available virtual memory active                      10716644
Total available virtual memory inactive                     3488756
Total available swap memory                                       0
Total available swap memory free                                  0
Total available memory that needs to be written to disk or swap  20
Total available memory which is actively being written back to
the disk                                                          0
Total Non-file backed pages mapped into userspace page tables 10483848
Total available memory that has been mapped (mmap)          3430036
Total available memory in the slab (kernel data structures)  102364
Total available memory part of slab, that might be reclaimed,
such as caches                                                52820
Total available memory part of slab that cannot be reclaimed on memory
pressure                                                      49544
Total available memory dedicated to lowest level page tables 120400
Total available memory NFS pages sent to the server, but not yet
committed to stable storage                                       0
Total available memory block device bounce buffers                0
Total available memory used by FUSE for temporary writeback buffers 0
Total available memory commit limit                        16407440
Total available memory presently allocated                 20186476
Total available virtual memory                 18446744073709551615
Total available virtual memory used                           69400
Largest available virtual memory chunk         18446744073709473688
```

```
Total available huge tlb pages                                           0
Total available huge tlb pages free                                      0
Total available huge tlb pages reserved                                  0
Total available huge tlb pages surplus                                   0
Total available huge tlb page size                                    2048

system statistics node-level memory-stats 11 3
Average 1 minute csm memory utilization                                  0
Average 5 minute csm memory utilization                                  0
Average 15 minute csm memory utilization                                 0
Average 1 minute ssm memory utilization                                 21
Average 5 minute ssm memory utilization                                 21
Average 15 minute ssm memory utilization                                21
Average 1 minute mcm memory utilization                                  0
Average 5 minute mcm memory utilization                                  0
Average 15 minute mcm memory utilization                                 0
Total available memory                                            32813984
Total available memory free                                       25742812
Total available memory in buffer cache                              139620
Total available memory in page cache minus swap cache               254152
Total available memory in swap cache                                     0
Total available virtual memory active                              3707048
Total available virtual memory inactive                             160192
Total available swap memory                                              0
Total available swap memory free                                         0
Total available memory that needs to be written to disk or swap          8
Total available memory which is actively being written back to
the disk                                                                 0
Total Non-file backed pages mapped into userspace
page tables                                                        3479396
Total available memory that has been mapped (mmap)                  146976
Total available memory in the slab (kernel data structures)          68704
Total available memory part of slab, that might be reclaimed,
such as caches                                                       32800
Total available memory part of slab that cannot be reclaimed on memory
pressure                                                             35904
Total available memory dedicated to lowest level page tables         18784
Total available memory NFS pages sent to the server, but not yet
committed to stable storage                                              0
Total available memory block device bounce buffers                       0
Total available memory used by FUSE for temporary writeback buffers  0
Total available memory commit limit                               14870992
Total available memory presently allocated                         4660384
Total available virtual memory                        18446744073709551615
```

```
Total available virtual memory used                              69400
Largest available virtual memory chunk            18446744073709475276
Total available huge tlb pages                                    1500
Total available huge tlb pages free                                  0
Total available huge tlb pages reserved                              0
Total available huge tlb pages surplus                               0
Total available huge tlb page size                                2048

system statistics node-level memory-stats 11 7
Average 1 minute csm memory utilization                              0
Average 5 minute csm memory utilization                              0
Average 15 minute csm memory utilization                             0
Average 1 minute ssm memory utilization                              0
Average 5 minute ssm memory utilization                              0
Average 15 minute ssm memory utilization                             0
Average 1 minute mcm memory utilization                             26
Average 5 minute mcm memory utilization                             26
Average 15 minute mcm memory utilization                            26
Total available memory                                         8042380
Total available memory free                                    5893432
Total available memory in buffer cache                          279536
Total available memory in page cache minus swap cache          1341344
Total available memory in swap cache                                 0
Total available virtual memory active                          3154424
Total available virtual memory inactive                         292144
Total available swap memory                                          0
Total available swap memory free                                     0
Total available memory that needs to be written to disk or swap    116
Total available memory which is actively being written back to
the disk                                                             0
Total Non-file backed pages mapped into userspace page tables  1825276
Total available memory that has been mapped (mmap)              162492
Total available memory in the slab (kernel data structures)     162776
Total available memory part of slab, that might be reclaimed,
such as caches                                                  111012
Total available memory part of slab that cannot be reclaimed
on memory pressure                                               51764
Total available memory dedicated to lowest level page tables     29408
Total available memory NFS pages sent to the server,
but not yet committed to stable storage                              0
Total available memory block device bounce buffers                   0
Total available memory used by FUSE for temporary writeback buffers  0
Total available memory commit limit                            4021188
Total available memory presently allocated                     3597436
```

```
Total available virtual memory                      18446744073709551615
Total available virtual memory used                                29472
Largest available virtual memory chunk              18446744073709513184
Total available huge tlb pages                                         0
Total available huge tlb pages free                                    0
Total available huge tlb pages reserved                                0
Total available huge tlb pages surplus                                 0
Total available huge tlb page size                                  2048

system statistics cpu-level cpu-stats

                   AVG  AVG  AVG    AVG  AVG  AVG    AVG  AVG  AVG
                   1MIN 5MIN 15MIN  1MIN 5MIN 15MIN  1MIN 5MIN 15MIN
                   CSM  CSM  CSM    SSM  SSM  SSM    MCM  MCM  MCM
                   CPU  CPU  CPU    CPU  CPU  CPU    CPU  CPU  CPU
CHASSIS SLOT CPU   UTIL UTIL UTIL   UTIL UTIL UTIL   UTIL UTIL UTIL
---------------------------------------------------------------------
11      1    1     5    5    5      0    0    0      0    0    0
11      3    1     0    0    0      0    0    0      0    0    0
11      7    1     0    0    0      0    0    0      2    2    2
system statistics cpu-level memory-stats 11 1 1
Average 1 minute csm memory utilization                               43
Average 5 minute csm memory utilization                               43
Average 15 minute csm memory utilization                              43
Average 1 minute ssm memory utilization                                0
Average 5 minute ssm memory utilization                                0
Average 15 minute ssm memory utilization                               0
Average 1 minute mcm memory utilization                                0
Average 5 minute mcm memory utilization                                0
Average 15 minute mcm memory utilization                               0
Total available memory                                          32814880
Total available memory free                                     18448240
Total available memory in buffer cache                            143548
Total available memory in page cache minus swap cache            3577692
Total available memory in swap cache                                   0
Total available virtual memory active                           10716644
Total available virtual memory inactive                          3488756
Total available swap memory                                            0
Total available swap memory free                                       0
Total available memory that needs to be written to disk or swap       20
Total available memory which is actively being written back to
the disk                                                               0
Total Non-file backed pages mapped into userspace page tables 10483848
Total available memory that has been mapped (mmap)               3430036
Total available memory in the slab (kernel data structures)       102364
```

```
Total available memory part of slab, that might be reclaimed,
such as caches                                                    52820
Total available memory part of slab that cannot be reclaimed
on memory pressure                                                49544
Total available memory dedicated to lowest level page tables    120400
Total available memory NFS pages sent to the server, but not yet
committed to stable storage                                           0
Total available memory block device bounce buffers                    0
Total available memory used by FUSE for temporary writeback buffers  0
Total available memory commit limit                            16407440
Total available memory presently allocated                     20186476
Total available virtual memory                     18446744073709551615
Total available virtual memory used                               69400
Largest available virtual memory chunk             18446744073709473688
Total available huge tlb pages                                        0
Total available huge tlb pages free                                   0
Total available huge tlb pages reserved                               0
Total available huge tlb pages surplus                                0
Total available huge tlb page size                                 2048
system statistics cpu-level memory-stats 11 3 1
Average 1 minute csm memory utilization                               0
Average 5 minute csm memory utilization                               0
Average 15 minute csm memory utilization                              0
Average 1 minute ssm memory utilization                              21
Average 5 minute ssm memory utilization                              21
Average 15 minute ssm memory utilization                             21
Average 1 minute mcm memory utilization                               0
Average 5 minute mcm memory utilization                               0
Average 15 minute mcm memory utilization                              0
Total available memory                                         32813984
Total available memory free                                    25742812
Total available memory in buffer cache                           139620
Total available memory in page cache minus swap cache            254152
Total available memory in swap cache                                  0
Total available virtual memory active                           3707048
Total available virtual memory inactive                          160192
Total available swap memory                                           0
Total available swap memory free                                      0
Total available memory that needs to be written to disk or swap       8
Total available memory which is actively being written back to
the disk                                                              0
Total Non-file backed pages mapped into userspace page tables  3479396
Total available memory that has been mapped (mmap)               146976
Total available memory in the slab (kernel data structures)       68704
```

```
Total available memory part of slab, that might be reclaimed,
such as caches                                                    32800
Total available memory part of slab that cannot be reclaimed
on memory pressure                                                35904
Total available memory dedicated to lowest level page tables     18784
Total available memory NFS pages sent to the server, but not yet
committed to stable storage                                           0
Total available memory block device bounce buffers                    0
Total available memory used by FUSE for temporary writeback buffers  0
Total available memory commit limit                            14870992
Total available memory presently allocated                      4660384
Total available virtual memory                     18446744073709551615
Total available virtual memory used                               69400
Largest available virtual memory chunk             18446744073709475276
Total available huge tlb pages                                     1500
Total available huge tlb pages free                                   0
Total available huge tlb pages reserved                               0
Total available huge tlb pages surplus                                0
Total available huge tlb page size                                 2048
system statistics cpu-level memory-stats 11 7 1
Average 1 minute csm memory utilization                               0
Average 5 minute csm memory utilization                               0
Average 15 minute csm memory utilization                              0
Average 1 minute ssm memory utilization                               0
Average 5 minute ssm memory utilization                               0
Average 15 minute ssm memory utilization                              0
Average 1 minute mcm memory utilization                              26
Average 5 minute mcm memory utilization                              26
Average 15 minute mcm memory utilization                             26
Total available memory                                          8042380
Total available memory free                                     5893432
Total available memory in buffer cache                           279536
Total available memory in page cache minus swap cache           1341344
Total available memory in swap cache                                  0
Total available virtual memory active                           3154424
Total available virtual memory inactive                          292144
Total available swap memory                                           0
Total available swap memory free                                      0
Total available memory that needs to be written to disk or swap     116
Total available memory which is actively being written back to
the disk                                                              0
Total Non-file backed pages mapped into userspace page tables   1825276
Total available memory that has been mapped (mmap)               162492
Total available memory in the slab (kernel data structures)      162776
```

```
Total available memory part of slab, that might be reclaimed,
such as caches                                                        111012
Total available memory part of slab that cannot be reclaimed
on memory pressure                                                     51764
Total available memory dedicated to lowest level page tables           29408
Total available memory NFS pages sent to the server, but not yet
committed to stable storage                                                0
Total available memory block device bounce buffers                         0
Total available memory used by FUSE for temporary writeback buffers       0
Total available memory commit limit                                  4021188
Total available memory presently allocated                           3597436
Total available virtual memory                          18446744073709551615
Total available virtual memory used                                    29472
Largest available virtual memory chunk                  18446744073709513184
Total available huge tlb pages                                             0
Total available huge tlb pages free                                        0
Total available huge tlb pages reserved                                    0
Total available huge tlb pages surplus                                     0
Total available huge tlb page size                                      2048
```

```
system statistics core-level cpu-stats
                            AVG    AVG    AVG     AVG    AVG    AVG    AVG    AVG    AVG
                            1MIN   5MIN   15MIN   1MIN   5MIN   15MIN  1MIN   5MIN   15MIN
                            CSM    CSM    CSM     SSM    SSM    SSM    MCM    MCM    MCM
                            CPU    CPU    CPU     CPU    CPU    CPU    CPU    CPU    CPU
CHASSIS SLOT CPU  CORE UTIL   UTIL   UTIL    UTIL   UTIL   UTIL   UTIL   UTIL   UTIL
------------------------------------------------------------------------------------------
11      1    1    1    5      5      5       0      0      0      0      0      0
11      1    1    2    4      5      4       0      0      0      0      0      0
11      1    1    3    6      5      5       0      0      0      0      0      0
11      1    1    4    4      4      4       0      0      0      0      0      0
11      1    1    5    5      5      4       0      0      0      0      0      0
11      1    1    6    5      5      4       0      0      0      0      0      0
11      1    1    7    3      5      5       0      0      0      0      0      0
11      1    1    8    4      4      5       0      0      0      0      0      0
11      3    1    1    0      0      0       2      2      1      0      0      0
11      3    1    2    0      0      0       2      2      1      0      0      0
11      3    1    3    0      0      0       0      0      1      0      0      0
11      3    1    4    0      0      0       4      2      1      0      0      0
11      3    1    5    0      0      0       0      0      0      0      0      0
11      3    1    6    0      0      0       1      1      0      0      0      0
11      3    1    7    0      0      0       0      0      0      0      0      0
11      3    1    8    0      0      0       0      0      0      0      0      0
11      3    1    9    0      0      0       0      0      0      0      0      0
11      3    1    10   0      0      0       0      0      0      0      0      0
11      3    1    11   0      0      0       0      0      0      0      0      0
11      3    1    12   0      0      0       0      0      0      0      0      0
11      3    1    13   0      0      0       0      0      0      0      0      0
11      3    1    14   0      0      0       0      0      0      0      0      0
11      3    1    15   0      0      0       0      0      0      0      0      0
11      3    1    16   0      0      0       0      0      0      0      0      0
```

1.4.5 Using Log Files

You can retrieve log files from the MCC via the Linux shell for analysis. However, you can also view the non-packet capture files in */var/log/eventlog.lnk* from the CLI. Navigate to the folder and press Tab to display a list of available filenames.

Note: *Customized logs/debugs are stored by default in the eventlog directory.*

```
an3000-mcm-slot7cpu1# show logs logfile <name>
```

TOS logs, which are binary by default, can also be viewed when archived from the Linux shell by unzipping the file and using the logconvert utility to convert the binary file to ASCII.

```
root@an3000-mcm-slot8cpu1:~#  cp tos20150419133044_33.bin.gz /tmp/
root@an3000-mcm-slot8cpu1:~#  cd /tmp/
root@an3000-mcm-slot8cpu1:~#  gunzip tos20150419133044_33.bin.gz
root@an3000-mcm-slot8cpu1:~#  logconvert tos20150419133044_33.bin
root@an3000-mcm-slot8cpu1:~#  cat tos20150419133044_33.bin.txt
```

1.5 GGSN/PGW Commands

Use the commands and parameters in this section to view gateway information and related statistics.

1.5.1 Call Performance Tables

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-
level call-performance pgw <table-name>
current-sessions                                        5
bearers                                                 5
create-session-attempts                                 41868
create-session-accepts                                  38943
create-session-rejects                                  2925
create-session-rejects-malformed-requests               0
create-session-rejects-unknown-apn                      1210
create-session-rejects-resource-unavailable             0
create-session-rejects-service-not-supported            0
create-session-rejects-system-failure                   1715
create-session-rejects-misc                             0
delete-session-attempts                                 25249
delete-session-accepts                                  25249
delete-session-rejects                                  0
ue-initiated-dedicated-bearer-activation-attempts       0
```

```
ue-initiated-dedicated-bearer-activation-accepts        0
ue-initiated-dedicated-bearer-activation-rejects        0
pgw-initiated-dedicated-bearer-activation-attempts      0
pgw-initiated-dedicated-bearer-activation-accepts       0
pgw-initiated-dedicated-bearer-activation-rejects       0
mme-initiated-dedicated-bearer-deactivation-attempts    0
mme-initiated-dedicated-bearer-deactivation-accepts     0
mme-initiated-dedicated-bearer-deactivation-rejects     0
ue-initiated-dedicated-bearer-deactivation-attempts     0
ue-initiated-dedicated-bearer-deactivation-accepts      0
ue-initiated-dedicated-bearer-deactivation-rejects      155
pgw-initiated-dedicated-bearer-deactivation-attempts    0
pgw-initiated-dedicated-bearer-deactivation-accepts     0
pgw-initiated-dedicated-bearer-deactivation-rejects     0
modify-bearer-command-attempts                          0
modify-bearer-command-rejects                           0
update-bearer-attempts                                  999
update-bearer-accepts                                   88
update-bearer-rejects                                   882
modify-bearer-attempts                                  1019633
modify-bearer-accepts                                   1019570
modify-bearer-rejects                                   0
create-pdp-context-attempts                             41147
create-pdp-context-accepts                              41006
create-pdp-context-rejects                              141
create-pdp-context-rejects-malformed-requests           0
create-pdp-context-rejects-unknown-apn                  0
create-pdp-context-rejects-resource-unavailable         0
create-pdp-context-rejects-service-not-supported        0
create-pdp-context-rejects-system-failure               140
create-pdp-context-rejects-misc                         1
delete-pdp-context-attempts                             41718
delete-pdp-context-accepts                              41718
delete-pdp-context-rejects                              0
ue-initiated-secondary-pdp-activation-attempts          0
ue-initiated-secondary-pdp-activation-accepts           0
ue-initiated-secondary-pdp-activation-rejects           0
ggsn-initiated-secondary-pdp-activation-attempts        0
ggsn-initiated-secondary-pdp-activation-accepts         0
ggsn-initiated-secondary-pdp-activation-rejects         0
sgsn-initiated-secondary-pdp-deactivation-attempts      0
sgsn-initiated-secondary-pdp-deactivation-accepts       0
sgsn-initiated-secondary-pdp-deactivation-rejects       0
```

```
ue-initiated-secondary-pdp-deactivation-attempts        0
ue-initiated-secondary-pdp-deactivation-accepts         0
ue-initiated-secondary-pdp-deactivation-rejects         172
ggsn-initiated-secondary-pdp-deactivation-attempts      0
ggsn-initiated-secondary-pdp-deactivation-accepts       0
ggsn-initiated-secondary-pdp-deactivation-rejects       0
sgsn-initiated-update-pdp-attempts                      19570
sgsn-initiated-update-pdp-accepts                       19330
sgsn-initiated-update-pdp-rejects                       0
ggsn-initiated-update-pdp-attempts                      7
ggsn-initiated-update-pdp-accepts                       6
ggsn-initiated-update-pdp-rejects                       1
local-close-attempts                                    12917
local-close-attempts-accepted                           12917
pcrf-initiated-close-attempts                           63
ocs-initiated-close-attempts                            0
idle-session-duration-exceeded-closes                   9456
maximal-session-duration-exceeded-closes                0
operator-initiated-pdn-session-closes                   6
active-sessions                                         2
idle-sessions                                           3
interval-peak-sessions                                  10961
interval-peak-active-sessions                           3770
interval-peak-idle-sessions                             7690
default-bearers                                         5
dedicated-bearers                                       0
interval-peak-bearers                                   10961
interval-peak-default-bearers                           10961
interval-peak-dedicated-bearers                         0
local-close-dhcp-release-attempts                       0
trace-session-activations                               0
trace-session-activation-accepts                        0
trace-session-deactivations                             0
trace-session-deactivation-accepts                      0
local-close-dhcp-timeout-attempts                       0
local-close-dhcp-lease-expiry-attempts                  0
local-close-radius-disconnect-closes                    0
current-sessions-rat-type-utran                         1
current-sessions-rat-type-geran                         0
current-sessions-rat-type-wlan                          0
current-sessions-rat-type-gan                           0
current-sessions-rat-type-hspa-evolution                0
current-sessions-rat-type-e-utran                       4
```

```
current-sessions-rat-type-virtual                        0
current-sessions-using-gz                                0
current-unique-subscribers                               5
current-sessions-with-co-located-peer                    0
current-ipv4-pdn-sessions                                5
current-ipv6-pdn-sessions                                0
current-dual-stack-pdn-sessions                          0
number-of-rat-session-handovers                          11332
number-of-sessions-setup-with-default-policy             1057
number-of-active-sessions-with-default-policy            0
number-of-active-gx-sessions-awaiting-pcrf-rejoin        0
number-of-successful-pcrf-rejoin                         888
number-of-gx-sessions-terminated-exceeding-reserve-usage 0
local-close-attempts-duplicate-session                   789
number-of-control-teid-lookup-failures                   1573
number-of-error-indications-processed                    200
local-close-received-error-indication-attempts           200
dedicated-bearer-deletion-due-to-error-indication        0
local-close-attempts-teid-collision                      0
change-notification-attempts                             0
change-notification-accepts                              0
change-notification-rejects                              0
change-notification-rejects-malformed-requests           0
change-notification-rejects-unknown-apn                  0
change-notification-rejects-resource-unavailable         0
change-notification-rejects-service-not-supported        0
change-notification-rejects-system-failure               0
change-notification-rejects-misc                         0
change-notification-req-drop                             0
num-gtpp-cdr-sent                                        528708
num-gtpp-cdr-resp-success                                528705
num-gtpp-cdr-resp-failure                                0
num-gtpp-cdr-resp-timeout                                3
bearers-setup-success                                    79949
bearers-setup-fail                                       1856
current-bearers-qci1                                     0
current-bearers-qci2                                     0
current-bearers-qci3                                     0
current-bearers-qci4                                     0
current-bearers-qci5                                     0
current-bearers-qci6                                     0
current-bearers-qci7                                     0
current-bearers-qci8                                     5
```

```
current-bearers-qci9                              0
current-bearers-qci-other                         0
bearers-setup-success-qci1                        0
bearers-setup-success-qci2                        0
bearers-setup-success-qci3                        0
bearers-setup-success-qci4                        0
bearers-setup-success-qci5                        0
bearers-setup-success-qci6                        0
bearers-setup-success-qci7                        0
bearers-setup-success-qci8                        79949
bearers-setup-success-qci9                        0
bearers-setup-success-qci-other                   0
bearers-setup-fail-qci1                           0
bearers-setup-fail-qci2                           0
bearers-setup-fail-qci3                           0
bearers-setup-fail-qci4                           0
bearers-setup-fail-qci5                           0
bearers-setup-fail-qci6                           0
bearers-setup-fail-qci7                           0
bearers-setup-fail-qci8                           1856
bearers-setup-fail-qci9                           0
bearers-setup-fail-qci-other                      0
saegw-pgw-current-sessions                        0
saegw-pgw-current-bearers                         0
saegw-pgw-bearers-setup-success                   14255
saegw-pgw-bearers-setup-fail                      599
saegw-pgw-current-bearers-qci1                    0
saegw-pgw-current-bearers-qci2                    0
saegw-pgw-current-bearers-qci3                    0
saegw-pgw-current-bearers-qci4                    0
saegw-pgw-current-bearers-qci5                    0
saegw-pgw-current-bearers-qci6                    0
saegw-pgw-current-bearers-qci7                    0
saegw-pgw-current-bearers-qci8                    0
saegw-pgw-current-bearers-qci9                    0
saegw-pgw-current-bearers-qci-other               0
saegw-pgw-bearers-setup-success-qci1              0
saegw-pgw-bearers-setup-success-qci2              0
saegw-pgw-bearers-setup-success-qci3              0
saegw-pgw-bearers-setup-success-qci4              0
saegw-pgw-bearers-setup-success-qci5              0
saegw-pgw-bearers-setup-success-qci6              0
saegw-pgw-bearers-setup-success-qci7              0
```

```
saegw-pgw-bearers-setup-success-qci8              14255
saegw-pgw-bearers-setup-success-qci9              0
saegw-pgw-bearers-setup-success-qci-other         0
saegw-pgw-bearers-setup-fail-qci1                 0
saegw-pgw-bearers-setup-fail-qci2                 0
saegw-pgw-bearers-setup-fail-qci3                 0
saegw-pgw-bearers-setup-fail-qci4                 0
saegw-pgw-bearers-setup-fail-qci5                 0
saegw-pgw-bearers-setup-fail-qci6                 0
saegw-pgw-bearers-setup-fail-qci7                 0
saegw-pgw-bearers-setup-fail-qci8                 599
saegw-pgw-bearers-setup-fail-qci9                 0
saegw-pgw-bearers-setup-fail-qci-other            0
local-close-await-coa-timeout-attempts            0
l2tp-gateway-session-delete-attempts              0
uplink-packets                                    148853580
uplink-bytes                                      28860983953
uplink-drop-packets                               109525
uplink-drop-bytes                                 12997621
downlink-packets                                  183123266
downlink-bytes                                    166218743634
downlink-drop-packets                             31489
downlink-drop-bytes                               43876356
uplink-suspend-drop-packets                       0
uplink-suspend-drop-bytes                         0
downlink-suspend-drop-packets                     0
downlink-suspend-drop-bytes                       0
num-suspend-notification                          0
num-suspend-notification-accepts                  0
num-suspend-notification-rejects                  0
num-resume-notification                           0
num-resume-notification-accepts                   0
num-resume-notification-rejects                   0
discontinuity-timestamp                           1427341689
```

1.5.2 Peer Chassis-level GTP Statistics Table

Use the **gtp peer PGW** parameter to display GTP performance statistics by peer.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics grid-level
gtp peer PGW
gateway statistics grid-level gtp peer PGW
create-session-request-decode-success                        0
create-session-request-encode-success                        0
```

```
create-session-request-decode-failure                                      0
create-session-request-encode-failure                                      0
create-session-response-decode-success                                     0
create-session-response-encode-success                                     0
create-session-response-decode-failure                                     0
create-session-response-encode-failure                                     0
delete-session-request-decode-success                                      118
delete-session-request-encode-success                                      0
delete-session-request-decode-failure                                      0
delete-session-request-encode-failure                                      0
delete-session-response-decode-success                                     0
delete-session-response-encode-success                                     115
delete-session-response-decode-failure                                     0
delete-session-response-encode-failure                                     0
modify-bearer-request-decode-success                                       2896
modify-bearer-request-encode-success                                       0
modify-bearer-request-decode-failure                                       0
modify-bearer-request-encode-failure                                       0
modify-bearer-response-decode-success                                      0
modify-bearer-response-encode-success                                      2875
modify-bearer-response-decode-failure                                      0
modify-bearer-response-encode-failure                                      0
create-bearer-request-decode-success                                       0
create-bearer-request-encode-success                                       0
create-bearer-request-decode-failure                                       0
create-bearer-request-encode-failure                                       0
create-bearer-response-decode-success                                      0
create-bearer-response-encode-success                                      0
create-bearer-response-decode-failure                                      0
create-bearer-response-encode-failure                                      0
delete-bearer-request-decode-success                                       0
delete-bearer-request-encode-success                                       0
delete-bearer-request-decode-failure                                       0
delete-bearer-request-encode-failure                                       0
delete-bearer-response-decode-success                                      0
delete-bearer-response-encode-success                                      0
delete-bearer-response-decode-failure                                      0
delete-bearer-response-encode-failure                                      0
bearer-resource-command-decode-success                                     0
bearer-resource-command-encode-success                                     0
bearer-resource-command-decode-failure                                     0
bearer-resource-command-encode-failure                                     0
bearer-resource-failure-indication-decode-success                          0
bearer-resource-failure-indication-encode-success                          0
bearer-resource-failure-indication-decode-failure                          0
bearer-resource-failure-indication-encode-failure                          0
modify-bearer-command-decode-success                                       0
modify-bearer-command-encode-success                                       0
modify-bearer-command-decode-failure                                       0
```

```
modify-bearer-command-encode-failure                                    0
modify-bearer-failure-indication-decode-success                         0
modify-bearer-failure-indication-encode-success                         0
modify-bearer-failure-indication-decode-failure                         0
modify-bearer-failure-indication-encode-failure                         0
update-bearer-request-decode-success                                    0
update-bearer-request-encode-success                                    0
update-bearer-request-decode-failure                                    0
update-bearer-request-encode-failure                                    0
update-bearer-response-decode-success                                   0
update-bearer-response-encode-success                                   0
update-bearer-response-decode-failure                                   0
update-bearer-response-encode-failure                                   0
downlink-data-notification-decode-success                               0
downlink-data-notification-encode-success                               0
downlink-data-notification-decode-failure                               0
downlink-data-notification-encode-failure                               0
downlink-data-notification-ack-decode-success                           0
downlink-data-notification-ack-encode-success                           0
downlink-data-notification-ack-decode-failure                           0
downlink-data-notification-ack-encode-failure                           0
downlink-data-notification-failure-indication-decode-success            0
downlink-data-notification-failure-indication-encode-success            0
downlink-data-notification-failure-indication-decode-failure            0
downlink-data-notification-failure-indication-encode-failure            0
release-access-bearers-request-decode-success                           0
release-access-bearers-request-encode-success                           0
release-access-bearers-request-decode-failure                           0
release-access-bearers-request-encode-failure                           0
release-access-bearers-response-decode-success                          0
release-access-bearers-response-encode-success                          0
release-access-bearers-response-decode-failure                          0
release-access-bearers-response-encode-failure                          0
change-notification-request-decode-success                              0
change-notification-request-encode-success                              0
change-notification-request-decode-failure                              0
change-notification-request-encode-failure                              0
change-notification-response-decode-success                             0
change-notification-response-encode-success                             0
change-notification-response-decode-failure                             0
change-notification-response-encode-failure                             0
create-pdp-context-request-decode-success                               0
create-pdp-context-request-encode-success                               0
create-pdp-context-request-decode-failure                               0
create-pdp-context-request-encode-failure                               0
create-pdp-context-response-decode-success                              0
create-pdp-context-response-encode-success                              0
create-pdp-context-response-decode-failure                              0
create-pdp-context-response-encode-failure                              0
```

```
delete-pdp-context-request-decode-success                              0
delete-pdp-context-request-encode-success                              0
delete-pdp-context-request-decode-failure                              0
delete-pdp-context-request-encode-failure                              0
delete-pdp-context-response-decode-success                             0
delete-pdp-context-response-encode-success                             2
delete-pdp-context-response-decode-failure                             0
delete-pdp-context-response-encode-failure                             0
update-pdp-context-request-decode-success                              0
update-pdp-context-request-encode-success                              0
update-pdp-context-request-decode-failure                              0
update-pdp-context-request-encode-failure                              0
update-pdp-context-response-decode-success                             0
update-pdp-context-response-encode-success                             0
update-pdp-context-response-decode-failure                             0
update-pdp-context-response-encode-failure                             0
num-echo-requests-received                                             0
num-echo-request-decode-failure                                        0
num-echo-requests-sent                                                 0
num-echo-responses-received                                            0
num-echo-response-decode-failure                                       0
num-echo-responses-sent                                                0
num-trace-session-activation-decode-success                            0
num-trace-session-activation-encode-success                            0
num-trace-session-activation-decode-failure                            0
num-trace-session-activation-encode-failure                            0
num-trace-session-deactivation-decode-success                          0
num-trace-session-deactivation-encode-success                          0
num-trace-session-deactivation-decode-failure                          0
num-trace-session-deactivation-encode-failure                          0
num-create-indirect-data-forwarding-tunnel-request-decode-success      0
num-create-indirect-data-forwarding-tunnel-request-decode-failure      0
num-create-indirect-data-forwarding-tunnel-response-encode-success     0
num-delete-indirect-data-forwarding-tunnel-request-decode-success      0
num-delete-indirect-data-forwarding-tunnel-request-decode-failure      0
num-delete-indirect-data-forwarding-tunnel-response-encode-success     0
num-suspend-notification-decode-success                                0
num-suspend-notification-encode-success                                0
num-suspend-notification-decode-failure                                0
num-suspend-notification-encode-failure                                0
num-suspend-acknowledge-decode-success                                 0
num-suspend-acknowledge-encode-success                                 0
num-suspend-acknowledge-decode-failure                                 0
num-suspend-acknowledge-encode-failure                                 0
num-resume-notification-decode-success                                 0
num-resume-notification-encode-success                                 0
num-resume-notification-decode-failure                                 0
num-resume-notification-encode-failure                                 0
num-resume-acknowledge-decode-success                                  0
```

```
num-resume-acknowledge-encode-success                                 0
num-resume-acknowledge-decode-failure                                 0
num-resume-acknowledge-encode-failure                                 0
discontinuity-timestamp                                               1427361106
```

1.5.3 IMSI Manager Message Statistics

The IMSI manager task runs on the CSM and maintains a database of all IMSIs registered on the MCC. It streamlines handling of GTP-C packets to ensure the SGW/PGW/GGSN task remains the same during a session's lifetime. It also pulls load information for each task to evenly distribute session requests across the tasks.

Use the following command to display the IMSI manager task (IMSIMGR) statistics:

```
an3000-mcm-slot7cpu1# show zone default gateway debug-stats imsimgr
message-statistics
number-of-gtpc-messages-received                                      2
number-of-gtpc-messages-dropped-not-synced                            0
number-of-messages-dropped-no-gw                                      0
number-of-gtpc-messages-dropped-wrong-message-type                    0
number-of-gtpc-messages-dropped-decode-fail                           0
number-of-gtpc-messages-identified-by-sequence-number                 0
number-of-dhcp-messages-received                                      0
number-of-dhcp-message-received-no-existing-subscriber                0
number-of-dhcp-messages-dropped-not-synced                            0
number-of-dhcp-messages-dropped-no-imsi                               0
number-of-dhcp-messages-forwarded                                     0
number-of-ip-packets-received                                         0
number-of-ip-packets-dropped-not-synced                               0
number-of-ip-packets-dropped-no-imsi                                  0
number-of-ip-packets-dropped-no-subscriber                            0
number-of-gtpu-messages-received                                      0
number-of-gtpu-messages-dropped-not-synced                            0
number-of-gtpu-messages-dropped-unknown-type                          0
number-of-gtpu-messages-dropped-decode-failure                        0
number-of-gtpu-error-indications-received                             0
number-of-radius-messages-received                                    0
number-of-radius-messages-dropped-not-synced                          0
number-of-radius-messages-dropped-decode-fail                         0
number-of-security-policy-manager-messages-received                   0
number-of-security-policy-manager-messages-received-with-imsi          0
number-of-security-policy-manager-messages-dropped-not-synced         0
number-of-security-policy-manager-messages-dropped-no-identity        0
number-of-security-policy-manager-messages-dropped-no-pseudonym-identity 0
number-create-request-forwarded-no-peer-pgw                           0
number-create-request-dropped-previous-gw-overload                    0
number-create-request-dropped-send-batch-fail                         0
number-create-request-dropped-add-to-batch-fail                       0
number-change-notification-request-dropped-add-to-batch-fail          0
number-change-notification-response-dropped-add-to-batch-fail         0
```

```
number-change-notification-request-dropped-decode-fail                 0
number-change-notification-response-dropped-decode-fail                0
number-create-request-dropped-slb-tbl-empty                            0
number-create-request-dropped-sgw-cant-accept                          0
number-create-request-dropped-pgw-cant-accept                          0
number-create-request-added-to-batch                                   2
number-change-notification-request-pinned-sgw-rate-ctrl-dropped        0
number-change-notification-response-pinned-sgw-rate-ctrl-dropped       0
number-change-notification-request-dropped-no-gw                       0
number-change-notification-response-dropped-no-gw                      0
number-change-notification-request-dropped-sgw-cant-accept             0
number-change-notification-response-dropped-sgw-cant-accept            0
number-change-notification-request-dropped-pgw-cant-accept             0
number-change-notification-response-dropped-pgw-cant-accept            0
number-change-notification-request-pinned-pgw-rate-ctrl-dropped        0
number-change-notification-response-pinned-pgw-rate-ctrl-dropped       0
number-create-requests-forwarded                                       2
number-create-request-send-batch-successes                             2
number-create-request-send-batch-failures                              0
number-create-request-resend-batch-success                             0
number-create-request-resend-batch-failure                             0
number-create-request-unmatched-new-peer-pgw                           0
number-create-request-unmatched-peer-sgw                               0
number-create-request-unmatched-peer-pgw                               0
number-create-request-resend-failure-no-gw                             0
number-create-request-pinned-sgw-rate-ctrl-dropped                     0
number-create-request-pinned-pgw-rate-ctrl-dropped                     0
number-emergency-create-request-received                               0
number-emergency-create-request-dropped-add-to-batch-fail              0
number-sgw-imsi-in-gw-list                                             0
number-pgw-imsi-in-gw-list                                             0
number-sgw-imsi-new-gw-selected                                        2
number-pgw-imsi-new-gw-selected                                        0
number-pinned-sgw-selected-pgw                                         0
number-pinned-sgw-pinned-pgw                                           0
number-selected-sgw-pinned-pgw                                         0
number-selected-sgw-selected-pgw                                       2
number-of-diameter-request-to-send-to-preferred-gw                     0
number-of-diameter-reqest-received                                     0
number-of-diameter-request-dropped-decode-fail                         0
number-of-diameter-request-dropped-invalid-format                      0
number-of-diameter-request-dropped-invalid-key                         0
number-of-diameter-request-dropped-no-imsi                             0
number-of-diameter-request-dropped-subscriber-not-found                0
number-of-diameter-request-dropped-task-not-found                      0
number-of-diameter-request-sent-to-gw                                  0
number-of-diameter-request-add-to-batch-fail                           0
number-create-request-rejected-admission-control                       0
```

1.5.4 GTP Peer Status

Use the **gtp-peer-status** parameter to display a status update on the GTP peering between the MCC gateway (SGW/PGW) and its GTP peer.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status
```

```
gateway gtp-peer-status
                                                                                        PEER     PEER  PEER
                                                         NETWORK  LOCAL IP       GTP     RESTART  NODE  NODE   PEER
GATEWAY NAME       PEER IP ADDRESS                       CONTEXT  ADDRESS        VERSION COUNTER  LCI   OCI    STATE
--------------------------------------------------------------------------------------------------------------------
7895:2000::1-2152  2001:db9:0:1:20c:29ff:fe59:ec38               7895:2000::1   gtpUv1  0        0     0      unknown
SGW                2001:db9:0:1:20c:29ff:fe59:ec38       GN       2007:700::7    gtpv2   0        0     0      down
```

1.5.4.1 GTP Peer Status Filtering by Network Context

Prior to release 9.2, you can use the following command to filter GTP peer status by network-context. This command only displays the GTP peers which are configured on the specified network-context, and filters out all other network contexts.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status
<network-context>
```

From release 9.2, you cannot use the command above to filter by network-context. You must use the following command and include the gateway name and gateway peer IP in addition to the network-context.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status
<gateway-name> <peer-ip> <network-context>
```

1.5.5 Service Status Tables

Use the **service-status** parameter to display a snapshot of the health and utilization of the gateway.

```
an3000-mcm-slot7cpu1# show zone default gateway service-status
Possible completions:
  epdg         ePDG Service Status Table
  ggsn         GGSN Service Status Table
  gtpproxy     GTP_PROXY Service Status Table
  mcc-gigw     MCC Service Status Table
  pgw          PGW Service Status Table
  sgw          SGW Service Status Table
  userplane    User Plane Service Status Table
  wag          WAG Service Status Table
```

For example:

```
an3000-mcm-slot7cpu1# show zone default gateway service-status
```

```
gateway service-status pgw
                                                                                   NUMBER
        GEO        ACTIVE       NUMBER OF    STANDBY      NUMBER OF  NUMBER    OF
        REDUNDANT  OPERATIONAL  ACTIVE       OPERATIONAL  STANDBY    OF        ACTIVE
NAME    GROUP      STATUS       INSTANCES    STATUS       INSTANCES  SESSIONS  SESSIONS
-------------------------------------------------------------------------------------
PGW     up         32           up           32           5          3         2
```

1.5.6 SLB Load Status Table

Use the **slb-load-status** parameter to display the load distribution status across the gateways.

```
an3000-mcm-slot7cpu1# show zone default gateway slb-load-status
gateway slb-load-status
GATEWAY                                     LOAD      CURRENT ABSOLUTE
TYPE     NAME                   SERVICEID   STATE     LOAD    VALUE
--------------------------------------------------------------------
sgw      SGW_100_11_1_0_SGW_0       152     MinorLow  0       0
sgw      SGW_100_11_1_12_SGW_24     741     MinorLow  0       0
sgw      SGW_100_11_1_16_SGW_32     747     MinorLow  0       0
sgw      SGW_100_11_1_20_SGW_40     751     MinorLow  0       0
sgw      SGW_100_11_1_24_SGW_48     755     MinorLow  0       0
sgw      SGW_100_11_1_28_SGW_56     759     MinorLow  0       0
sgw      SGW_100_11_1_4_SGW_8       699     MinorLow  0       0
sgw      SGW_100_11_1_8_SGW_16      721     MinorLow  0       0
sgw      SGW_100_12_1_13_SGW_26     742     MinorLow  0       0
sgw      SGW_100_12_1_17_SGW_34     748     MinorLow  0       0
sgw      SGW_100_12_1_1_SGW_2       164     MinorLow  0       0
sgw      SGW_100_12_1_21_SGW_42     752     MinorLow  0       0
sgw      SGW_100_12_1_25_SGW_50     756     MinorLow  0       0
sgw      SGW_100_12_1_29_SGW_58     760     MinorLow  0       0
sgw      SGW_100_12_1_5_SGW_10      703     MinorLow  0       0
sgw      SGW_100_12_1_9_SGW_18      725     MinorLow  0       0
sgw      SGW_100_13_1_10_SGW_20     730     MinorLow  0       0
sgw      SGW_100_13_1_14_SGW_28     743     MinorLow  0       0
sgw      SGW_100_13_1_18_SGW_36     749     MinorLow  0       0
sgw      SGW_100_13_1_22_SGW_44     753     MinorLow  0       0
sgw      SGW_100_13_1_26_SGW_52     757     MinorLow  0       0
sgw      SGW_100_13_1_2_SGW_4       176     MinorLow  0       0
sgw      SGW_100_13_1_30_SGW_60     761     MinorLow  0       0
sgw      SGW_100_13_1_6_SGW_12      708     MinorLow  0       0
sgw      SGW_100_14_1_11_SGW_22     736     MinorLow  0       0
pgw      PDNGW_100_11_1_0_PDNGW_0    145    MinorLow  0       0
pgw      PDNGW_100_11_1_12_PDNGW_24  589    MinorLow  0       0
pgw      PDNGW_100_11_1_16_PDNGW_32  611    MinorLow  0       0
pgw      PDNGW_100_11_1_20_PDNGW_40  633    MinorLow  0       0
pgw      PDNGW_100_11_1_24_PDNGW_48  655    MinorLow  0       0
pgw      PDNGW_100_11_1_28_PDNGW_56  677    MinorLow  0       0
```

```
pgw     PDNGW_100_11_1_4_PDNGW_8    545       MinorLow  0       0
pgw     PDNGW_100_11_1_8_PDNGW_16   567       MinorLow  0       0
pgw     PDNGW_100_12_1_13_PDNGW_26  593       MinorLow  0       0
pgw     PDNGW_100_12_1_17_PDNGW_34  615       MinorLow  0       1
pgw     PDNGW_100_12_1_1_PDNGW_2    157       MinorLow  0       0
pgw     PDNGW_100_12_1_21_PDNGW_42  637       MinorLow  0       0
pgw     PDNGW_100_12_1_25_PDNGW_50  659       MinorLow  0       0
pgw     PDNGW_100_12_1_29_PDNGW_58  681       MinorLow  0       0
pgw     PDNGW_100_12_1_5_PDNGW_10   549       MinorLow  0       0
pgw     PDNGW_100_12_1_9_PDNGW_18   571       MinorLow  0       0
pgw     PDNGW_100_13_1_10_PDNGW_20  576       MinorLow  0       0
pgw     PDNGW_100_13_1_14_PDNGW_28  598       MinorLow  0       0
pgw     PDNGW_100_13_1_18_PDNGW_36  620       MinorLow  0       0
pgw     PDNGW_100_13_1_22_PDNGW_44  642       MinorLow  0       0
pgw     PDNGW_100_13_1_26_PDNGW_52  664       MinorLow  0       0
pgw     PDNGW_100_13_1_2_PDNGW_4    169       MinorLow  0       0
pgw     PDNGW_100_13_1_30_PDNGW_60  686       MinorLow  0       0
pgw     PDNGW_100_13_1_6_PDNGW_12   554       MinorLow  0       0
pgw     PDNGW_100_14_1_11_PDNGW_22  582       MinorLow  0       0
```

1.6 Network Context Commands

Networking of the MCC is based on the concept of a network context, used by Virtual Routing and Forwarding (VRF), which consists of multiple, independent routing instances/tables that are created on the MCC.

Table 1-3 MCC Network Contexts

Network Context Name	Description
GnGp-NC[a]	Interface to SGSN
Gx-NC	Interface to PCRF
Gy-NC	Interface to OCS
GiSGi-NC[a]	Interface to Internet
management-context	Supports OAM traffic, including EMS

[a] Bearer network contexts.

1.6.1 show network context GnGp-NC ip interface

```
an3000-mcm-slot7cpu1# show network-context GnGp-NC ip interface
ip interface GnGp-3-1
 IP Address           185.114.248.66
 IP Prefix Length     30
 IP Address           ""
 IP Prefix Length     ""
 OperStatus           UP
```

```
 TxV4Pkts             82
 TxV4Bytes            3940
 TxV4DropPkts         0
 TxV4DropBytes        0
 RxV4Pkts             6861
 RxV4Bytes            439074
 RxV4DropPkts         17
 RxV4DropBytes        796
 RxICMPv4Pkts         4
 TxICMPv4Pkts         0
 TxV6Pkts             4
 TxV6Bytes            316
 TxV6DropPkts         0
 TxV6DropBytes        0
 RxV6Pkts             0
 RxV6Bytes            0
 RxV6DropPkts         0
 RxV6DropBytes        0
 RxICMPv6Pkts         0
 TxICMPv6Pkts         0
 AclDropPkts          0
 AclDropBytes         0
 AclAllowPkts         0
 AclAllowBytes        0
ip interface GnGp-3-2
 IP Address           185.114.248.70
 IP Prefix Length     30
 IP Address           ""
 IP Prefix Length     ""
 OperStatus           UP
 TxV4Pkts             78
 TxV4Bytes            3588
 TxV4DropPkts         0
 TxV4DropBytes        0
 RxV4Pkts             6868
 RxV4Bytes            439410
 RxV4DropPkts         22
 RxV4DropBytes        1012
 RxICMPv4Pkts         5
 TxICMPv4Pkts         0
 TxV6Pkts             5
 TxV6Bytes            398
 TxV6DropPkts         0
 TxV6DropBytes        0
 RxV6Pkts             0
 RxV6Bytes            0
 RxV6DropPkts         0
 RxV6DropBytes        0
 RxICMPv6Pkts         0
```

```
 TxICMPv6Pkts        0
 AclDropPkts         0
 AclDropBytes        0
 AclAllowPkts        0
 AclAllowBytes       0
```

You can apply additional parameters to the IP interface command to display specific fields. For example:

```
an3000-mcm-slot7cpu1# show network-context GnGp-NC ip interface
oper-status ip interface
IP IF      OPER
NAME       STATUS
------------------
GnGp-3-1   UP
GnGp-3-2   UP

an3000-mcm-slot7cpu1# show network-context GiSGi-NC ip interface
oper-status ip interface
IP IF      OPER
NAME       STATUS
-------------------
GiSGi-3-1  UP
GiSGi-3-2  UP

an3000-mcm-slot7cpu1# show network-context GnGp-NC ip interface
[select rx-v4-bytes | select rx-v4-pkts | select tx-v4-bytes | select
tx-v4-pkts | select rx-v4-drop-bytes | select rx-v4-drop-pkts | select
tx-v4-drop-bytes | select tx-v4-drop-pkts ip interface-statistics]
ip interface
           TX                    RX
           TX     V4     TX V4   RX     V4     RX V4
IP IF      V4     TX V4  DROP    DROP   V4     RX V4    DROP  DROP
NAME       PKTS   BYTES  PKTS    BYTES  PKTS   BYTES    PKTS  BYTES
---------------------------------------------------------------------
GnGp-3-1   82     3940   0       0      6905   441890   19    888
GnGp-3-2   79     3634   0       0      6913   442290   22    1012

an3000-mcm-slot7cpu1# show network-context ip interface-statistics
network-context GiSGi-NC
ip interface-statistics GiSGi-3-1
 TxV4Pkts            90
 TxV4Bytes           4812
 TxV4DropPkts        0
 TxV4DropBytes       0
 RxV4Pkts            6725
 RxV4Bytes           430740
 RxV4DropPkts        32
 RxV4DropBytes       1484
```

```
RxICMPv4Pkts         28
TxICMPv4Pkts         0
TxV6Pkts             5
TxV6Bytes            398
TxV6DropPkts         0
```

1.6.2 show cluster node port-stats

The **show cluster node port-stats** is not a network context command, but it displays performance statistics on E/W (dataFastpath) and N/S (externalPort) communications on the IOM and WSM VMs.

```
an3000-mcm-slot7cpu1# show cluster node port-stats
cluster
                                                                                    RX          TX
                             RX                 RX PKTS   TX                 TX PKTS   RX   BYTES  TX   BYTES  UTIL
ID   ID   NAME               PKTS      RX BYTES DROPPED   PKTS     TX BYTES  DROPPED   PPS  PS     PPS  PS     PERCENT
------------------------------------------------------------------------------------------------------------------------
67   1    eth1-dataApps      6665      649385    0        6842     850680     0        0    0      0    0      0
     2    eth1-dataApps      359391    64667420  0        113110   10522543   0        4    456    3    291    0
     5    eth1-dataFastpath  114262    51776315  0        335333   104392384  0        1    104    2    247    0
          eth3-externalPort1 784343    105801902 0        646238   92701463   0        15   1170   12   849    0
          eth4-externalPort2 56789     4567766   0        0        0          0        1    97     0    0      0
          eth5-externalPort3 2         120       0        0        0          0        0    0      0    0      0
     6    eth1-dataFastpath  124872    54469110  0        149575   55763195   0        1    187    1    209    0
          eth3-externalPort1 691807    50471592  0        595070   41565461   0        14   1050   13   919    0
          eth4-externalPort2 57904     4659280   0        0        0          0        1    94     0    0      0
```

1.7 Proxy Commands

Use the commands and parameters in this section to view multi-protocol proxy cluster-level and node-level information and statistics.

1.7.1 Renamed HTTP Proxy Commands

In MCC release 8.4 and later, **http-proxy** was renamed to **multi-protocol-proxy** to encompass the following supported protocols: HTTP, TCP, HTTPS, UDP, and QUIC.

This change affects the following config and show commands:

- services http-proxy to services multi-protocol-proxy
- services crbn-profile <*name*> http-proxy to services crbn-profile <*name*> multi-protocol-proxy
- workflow data-profile <*name*> http-proxy to workflow data-profile <*name*> multi-protocol-proxy
- services http-proxy <*name*> http-ip-flow to services multi-protocol-proxy <*name*> ip-flow
- show services http-proxy to show services multi-protocol-proxy

1.7.2 multi-protocol-proxy connection

Use the following command to display information about the HTTP cluster proxy client and server connections:

```
an3000-mcm-slot7cpu1# show services multi-protocol-proxy statistics
cluster-level connection <chassis>
services multi-protocol-proxy statistics cluster-level connection 67
 Client Connections - Total                              2
 Client Connections - Currently in Use                   0
 Client Connections - Keep-Alive Requested               2
 Client Connections - Persistent                         2
 Client Connections - Reused                             1
 Client Connections - Pipelined                          0
 Client Connections - IPv4                               2
 Client Connections - IPv6                               0
 Server Connections - Total                              2
 Server Connections - Currently in Use                   0
 Server Connections - IPv4                               2
 Server Connections - IPv6                               0
 Server Connections - Currently Idle                     0
 Server Connections - Reused by Same Client              0
 Server Connections - Reused from Pool                   0
 Server Connections - Preferred IP Family Used           0
 Server Connections - Preferred IP Family Not Available  0
 Smpp Connections - Total                                0
 Smpp Connections - Currently in Use                     0
 Smpp Connections - IPv4                                 0
 Smpp Connections - IPv6                                 0
 Smpp Connections - Currently Idle                       0
 Smpp Connections - Reused by Same Client                0
 Smpp Connections - Reused                               0
 Smpp Connections -Connection Error                      0
 Smpp Connections - Bind Error                           0
 Smpp Connections - Send Error                           0
 Smpp Connections - Unbind Error                         0
 Server Connections - IP Transparent                     1
 Client Connections - Non-HTTP Message Received          0
 Client Connections - Request Receive Timeouts           0
 Client Connections - Reverse Handoffs                   0
 Client Connections - Failures                           0
 Fast Open - Attempts                                    0
 Fast Open - Successes                                   0
 Fast Open - Connection Initialization Failures          0
 Fast Open - Connection Already in Pending List          0
 Fast Open - Connection Already in Session               0
 Fast Open - Policy Denied                               0
```

```
Fast Open - Not Ready                                     0
Fast Open - Pending Connection List Full                  0
Fast Open - Current Pending Connections                   0
Fast Open - Pending Connection Idle Timeout               0
Discontinuity Timestamp                                   1561076905
```

1.7.3 multi-protocol-proxy summary

Proxy issues may be specific to a single ASM or a subset of the ASM VMs. Use the following command to check the health of each VM individually:

```
an3000-mcm-slot7cpu1# show services multi-protocol-proxy statistics
node-level summary <chassis> <slot>
```

```
services multi-protocol-proxy statistics node-level summary 125 5
 HTTP-Proxy Service - Attempts                0
 HTTP-Proxy Service - Successes               0
 HTTP-Proxy Service - Successfully Cached     0
 HTTP-Proxy Service - Aborts                  0
 HTTP-Proxy Service - Failures                0
 Client Requests - Total                      0
 Client Requests - Current                    0
 Client Requests - Parsed Successfully        0
 Client Requests - Incomplete                 0
 Client Requests - Processing Failed          0
 Client Requests - Pipeline Dropped           0
 Client Requests - Invalid                    0
 Client Requests - Total Bytes                0
 Client Responses - Total Bytes               0
 Client Sessions - Current                    0
 Server Requests - Total Bytes                0
 Server Responses - Total Bytes               0
 Service Time - Min (ms)                      0
 Service Time - Avg (ms)                      0
 Service Time - Max (ms)                      0
 Server Responses - Total                     0
 Server Responses - Min Time (ms)             0
 Server Responses - Avg Time (ms)             0
 Server Responses - Max Time (ms)             0
 Latency - Request Latency Average (ms)       0
 Latency - Response Latency Average (ms)      0
 Latency - Transaction Latency Average (ms)   0
 Discontinuity Timestamp                      1560362380
```

1.7.4 multi-protocol-proxy overload-control

Use the following command to check if the proxy has too much load:

```
an3000-mcm-slot7cpu1# show services multi-protocol-proxy statistics
cluster-level overload-control <chassis> updated

services multi-protocol-proxy statistics cluster-level overload-
control 125
 Overload Control - Connection Accepts 2013
 Overload Control - Connection Denies  0
 Discontinuity Timestamp               1427830262
```

1.8 IP Interface Packet Capture Utility

Use the following commands to perform a packet capture in pcapng format on a single IP interface configured in the system.

Note: *This command affects system performance and should only be used for short durations when troubleshooting.*

For example:

```
an3000-mcm-slot7cpu1# network-context
Possible completions:
GiSGi-NC GnGp-NC Gx-NC Gy-NC management-context

an3000-mcm-slot7cpu1# network-context GiSGi-NC ip-interface
Possible completions:
GiSGi-3-1 GiSGi-3-2

an3000-mcm-slot7cpu1# network-context GiSGi-NC ip-interface GiSGi-3-1
Possible completions:
startcapture        Enable packet capture
stopcapture         Disable packet capture

an3000-mcm-slot7cpu1# network-context GiSGi-NC ip-interface GiSGi-3-1
startcapture
Possible completions:
count               max number of packets to capture, default is 1000
duration            duration in seconds, default is 1
file-name           file name prefix
|                   Output modifiers
```

In the example highlighted text, the packet trace is run against a single IP interface configured in a single network context. This means you can run a packet trace on either a Gn or Gi network context from a single command execution, but not both. However, you can start a packet capture on a Gn IP-interface and start another parallel packet capture on a Gi IP-interface during the same period, but not in the same single line command.

The packet captures are stored in */var/log/eventlog.lnk* and can be retrieved using SCP/SFTP or EMS.

The filename format is <filenameprefix><YYYYMMDDHHMMSS>.pcap, for example, PktTr_20140922111150.pcapng. The file is saved as pcapng and is readable with Wireshark or similar packet analyzer software.

1.8.1 Packet Capture Considerations

In an MCC system with multiple active ingress/egress interfaces in each network context, traffic may be distributed among other interfaces, so the packet trace from a single interface may not represent the full traffic required.

Subscriber information (IMSI, etc.) is available on Gn side captures inside of the Create Session Request and similar GTP packets. From the Gn side, the payload is encapsulated in the GTP header and the source UE IP address is a pre-NAT private address.

Traffic captured on the Gi interface is post-NAT without the original private UE source IP address. No subscriber information is available on the Gi side traces.

1.9 Trace Logging

You can configure the MCC to log control plane information about a specific subscriber or subset of subscribers (for example, APN) up to a full debug 5 level.

The following command is an example of the required configuration:

```
an3000-mcm-slot7cpu1# infrastructure logging event-log policy TEST
log-type debug filter-field

Possible completions:

apn component componentlogid cpu gwname httpreqtype imei imsi
ipaddress msisdn npuflowid npunum npuslot portnumber service severity
slot teid vpn


infrastructure file-management profile eventlog IMSIDeb_
 purge-thr                    100
 purge-low-thr                0
 file-rotate-size-bytes       10485760
 file-transfer-enabled        disabled
 upload-frequency             1hour
```

```
 upload-offset-hour           0
 upload-offset-min            0
 file-rotate-frequency        1hour
 purge-file-older-than        30days
 purge-enabled                disabled
 compress-enabled             disabled
!
```

```
infrastructure logging event-log policy IMSIDeb
 admin-state              enabled
 log-type                 debug
 filter-field             imsi
  filter-value            <imsi>
   exclude                false
  !
 !
 filter-field             severity
  filter-value            Debug5
  !
 !
 sinks                    IMSIDeb
  type                    file
  file-name               IMSIDeb_
  file-format             ascii
 !
!
```

```
an3000-mcm-slot7cpu1# infrastructure logging event-log policy TEST
log-type protocol protocol-logging-filter-field
Possible completions:
destination-ip-address destination-portnumber imei imsi ip-address-
pair ip-protocol ipaddress msisdn packet-direction packet-type
protocol source-ip-address source-portnumber
```

```
an3000-mcm-slot7cpu1# infrastructure logging event-log policy TEST
log-type protocol protocol-logging-filter-field protocol filter-value
Possible completions:
dhcp diameter gtpp gtpv0 gtpv1 gtpv2 radius
```

```
infrastructure logging event-log policy GTPCapture
 admin-state enabled
 log-type protocol
 protocol-logging-filter-field protocol
  filter-value gtpv1
   filter-field interface
    filter-value gn
     exclude false
```

```
      !
      filter-value gp
       exclude false
      !
     !
    !
    filter-value gtpv2
     filter-field interface
      filter-value s4
       exclude false
      !
      filter-value s5
       exclude false
      !
      filter-value s8
       exclude false
      !
      filter-value s11
       exclude false
      !
     !
    !
   !
   sinks GTPCapture
    type file
    file-name GTPTrace_
    file-format pcapng
    !
   !
```

In the example highlighted text, the logging configuration supports both debug and protocol type event logs. The logs are then written to a local or remote disk for analysis. The logs are stored int */var/log/eventlog.lnk* and can be retrieved using SCP/SFTP or EMS.

The filename format is <filenameprefix><YYYYMMDDHHMMSS>.gz or .pcap, for example, tos20140919165835.txt.gz or GTPTrace _20140922111150.pcapng.

1.9.1 Trace Logging Considerations

The file management profile must be created before creating the event-log policy because every policy must reference a pre-existing profile. The profile does not need to be committed, but it should at least be entered into the pending configuration.

The log type can be either debug or protocol. If the debug log type is used, all traffic can be captured in binary or ASCII format and contains all flow information for the specified subscriber(s) given the parameters. For efficiency, binary format is recommended, and you can use the EMS log viewer to decode the log.

If the protocol log type is used, all traffic can be captured in ASCII or pcapng format. Pcapng format is readable with Wireshark for analysis.

Setting up event logs is a valuable troubleshooting tool for capturing specific control and/or data plane traffic that transits MCC, as well as system logs.

Note: *The information collected is subject to the configured criteria, and only control plane traffic is captured, not data plane. Also, while some logs can be configured for general use, using debug logs can affect system resources, so they should only be used for troubleshooting and then disabled.*

1.10 DPI Statistics

Use the following command to display a overview of Deep Packet Inspection (DPI) statistics:

```
an3000-mcm-slot7cpu1# show zone default deep-packet-inspection
statistics grid-level
deep-packet-inspection statistics grid-level general 1
 protocol-undetected-flows            259918
 detection-flows                      6779834
 detection-packets                    29205756
 detection-bytes                      5378670718
 decode-flows                         6779834
 decode-packets                       255842051
 decode-bytes                         155053953966
 protocol-undetected-packets          5045467
 protocol-undetected-bytes            2387156909
 detection-total-packets              312081744
 detection-total-bytes                190676236068
 decode-total-packets                 312081744
 decode-total-bytes                   190676236068
 no-dpi-flows                         7923117
 no-dpi-total-packets                 22259245
 no-dpi-total-bytes                   6476854861
 detection-uplink-packets             20289877
 detection-uplink-bytes               3391335882
 detection-downlink-packets           8915879
 detection-downlink-bytes             1987334836
 decode-uplink-packets                116264718
 decode-uplink-bytes                  18684391037
 decode-downlink-packets              139577333
 decode-downlink-bytes                136369562929
```

```
protocol-undetected-uplink-packets   2248227
protocol-undetected-uplink-bytes     739312028
protocol-undetected-downlink-packets 2797240
protocol-undetected-downlink-bytes   1647844881
detection-total-uplink-packets       138466421
detection-total-uplink-bytes         27614401916
detection-total-downlink-packets     173615323
detection-total-downlink-bytes       163061834152
decode-total-uplink-packets          138466421
decode-total-uplink-bytes            27614401916
decode-total-downlink-packets        173615323
decode-total-downlink-bytes          163061834152
no-dpi-total-uplink-packets          11222018
no-dpi-total-uplink-bytes            1497485662
no-dpi-total-downlink-packets        11037227
no-dpi-total-downlink-bytes          4979369199
amqp-flows                           54877714
amqp-uplink-packets                  882145547
amqp-uplink-bytes                    441255541
amqp-downlink-packets                619357447
amqp-downlink-bytes                  439871285
```

Use the following parameters to display DPI statistics for a specific category:

```
applications        DPI application statistics
business
conference
database
fileTransfer
gaming
general             DPI general statistics
im
mail
mobile
networkManagement
operatingSystems    DPI operating system statistics
p2p
remoteControl
socialNetworking
standard
streaming
tethering           DPI tethering statistics
tunnel
voip
```

```
web
|                    Output modifiers
```

You can specify specific applications within these categories. For example:

```
an3000-mcm-slot7cpu1# show zone default deep-packet-inspection
statistics grid-level applications 1 FaceTime-downlink-bytes
```

1.11 BGP Routing

The **SUPPORTZONE** and **SAEGW** network contexts use BGP as the primary routing method and form BGP relationships with adjacent CE/PE routers.

The `bgp router summary` command displays the health of the BGP peer relationships:

```
an3000-mcm-slot7cpu1# show ip-protocols bgp router <network-context>
summary
bgp router XXXX
summary entry
BGP router identifier 155.165.112.65, local AS number 65202
BGP table version is 9557
2 BGP AS-PATH entries
0 BGP community entries
6  Configured ebgp ECMP multipath: Currently set at 6
1  Configured ibgp ECMP multipath: Currently set at 1
Neighbor                 V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.30.33             4 65102  101814   95182     9557    0    0 6d14h50m        1
172.16.30.37             4 65102  101682   95147     9557    0    0 1d20h26m        1
172.16.30.41             4 65102  101812   95182     9557    0    0 6d14h50m        1
2606:ae00:2031:1008::    4 65102  101801   95171     9557    0    0 6d14h48m        0
2606:ae00:2031:1008::2   4 65102  101647   95109     9557    0    0 1d20h25m        0
2606:ae00:2031:1008::4   4 65102  101814   95182     9557    0    0 6d14h50m        0
2606:ae00:2031:1008::6   4 65102   92183   86275        0    0    0 never Active
```

```
an3000-mcm-slot7cpu1# show ip-protocols bgp router <network-context>
ipv4-ucast-af
ipv4-ucast-af
 route-table entry
BGP table version is 5736, local router ID is 107.77.163.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, l -
labeled, m - multipath
               S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*   m0.0.0.0/0        172.20.0.37             0             0 65101 20057 i
*   m                 172.20.0.45             0             0 65101 20057 i
*>  m                 172.20.0.33             0             0 65101 20057 i
*   m                 172.20.0.41             0             0 65101 20057 i
*>  107.77.163.1/32  127.0.0.2              1          32768 ?
*>  107.77.163.2/32  127.0.0.2              1          32768 ?
*>  107.77.163.3/32  127.0.0.2              1          32768 ?
*>  107.77.163.4/32  127.0.0.2              1          32768 ?
*>  107.77.163.5/32  127.0.0.2              1          32768 ?
Total number of prefixes 6
```

```
an3000-mcm-slot7cpu1# show ip-protocols bgp router <network-context>
ipv6-ucast-af
ipv6-ucast-af
 route-table entry
BGP table version is 5737, local router ID is 107.77.163.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, l -
labeled, m - multipath
              S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*   m::/0             2600:300:2000:3407::2(fe80::6664:9b0d:49b5:29c3)
                                             0             0 65101 20057 i
*   m                 2600:300:2000:3407::6(fe80::6664:9b0d:4bb5:29c7)
                                             0             0 65101 20057 i
*>  m                 2600:300:2000:3407::(fe80::6664:9b0d:48bc:2ac1)
                                             0             0 65101 20057 i
*   m                 2600:300:2000:3407::4(fe80::6664:9b0d:4abc:2ac5)
                                             0             0 65101 20057 i
*>  2600:387:0:900::/126
                    ::                                 32768 ?
*>  2600:387:0:900::1/128
                    ::1                      1         32768 ?
*>  2600:387:0:900::2/128
                    ::1                      1         32768 ?
*>  2600:387:0:900::3/128
                    ::1                      1         32768 ?
*>  2600:387:0:900::4/126
                    ::                                 32768 ?
*>  2600:387:0:900::4/128
                    ::1                      1         32768 ?
*>  2600:387:0:900::5/128
                    ::1                      1         32768 ?
```

1.12 Bidirectional Forwarding Detection

Bidirectional forwarding detection (BFD) allows for fast failure detection between the routing engine of BFD peered neighbors over multiple routing protocols (including static) with a minimum of overhead. BFD also improves failure detection time. However, it can also become a point of failure if there are configuration or software issues related to the BFD protocol.

1.12.1 show ip-protocols bfd

Use the **interfaces** parameter for the BFD command to display the status of each BFD interface based on the responses to echo packets sent at configured intervals over that interface.

```
an3000-mcm-slot7cpu1# show ip-protocols bfd instance <network-context>
interfaces
interfaces entry


Interface: XXXX_1_1_5  ifindex: 68 state:   UP
```

```
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_2_2_5  ifindex: 69 state:   UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_3_1_5  ifindex: 70 state:   UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_4_2_5  ifindex: 71 state:   UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3
```

Use the **sessions** parameter of the BFD command to display the status of the peering between the BFD neighbors and indicators of stability, along with the configured/default timers.

```
an3000-mcm-slot7cpu1# show ip-protocols bfd instance <network-context> sessions
sessions entry
Session Interface Index : 70          Session Index : 1
Lower Layer : IPv4                    Version : 1
Session Type : Single Hop             Session State : Up
Local Discriminator : 1               Local Address : 172.20.0.42/32
Remote Discriminator : 1469           Remote Address : 172.20.0.41/32
Local Port : 49152                    Remote Port : 3784
Options :

Diagnostics : None

Timers in Milliseconds :
Min Tx: 500             Min Rx: 500             Multiplier: 3
Neg Tx: 2000            Neg Rx: 2000            Neg detect mult: 3
Min echo Tx: 200        Min echo Rx: 100        Neg echo intrvl: 0
Storage type : 2
Sess down time : 00:00:00
Sess discontinue time : 00:00:00
Bfd GTSM Disabled
Bfd Authentication Disabled
UP Count : 0                            UPTIME : 7d20h53m
```

```
Protocol Client Info:
BGP-> Client ID: 7       Flags: 4
-----------------------------------------------------------
```

1.13 HTTP Transaction Data Records (HTDR)

For each transaction processed by the MCC multi-protocol proxy, an HTTP Transaction Data Record (HTDR) entry is created at transaction termination. This may be on flow termination or when a new HTTP GET transaction is allocated for a current flow (such as a new YouTube video chunk as part of the full stream). When MCC is sized correctly, there is no additional impact on system resources.

HTDRs are stored locally on the system at */public/logs/datarecord.lnk*. The filename format is <filename-prefix><YYYYMMDDHHMMSS>_<123>.csv, for example, HTDR20170919103140_123.csv. They are stored natively as CSV files and a data-template parser can be built.

The following example shows sample output for a single entry:

```
'1411147899,221,354','111113076999899,999930769998990,3076999899,21486
30528,226,10,5,6,1,2606ae00b8001b000000000000000001,,reseller','MVNO1,
0a000003,1024,4a7de048,80,0','2,http%3a%2f%2fwww.google.com%2f%3fgws_r
d%3dssl,,0,0,Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET
CLR
1.1.4322),,,203,0,203,0,0,1,0,0,0,9ba57302,6279,4a7de048,80,0','200,te
xt/html; charset=UTF-8,text/html; charset=UTF-
8,11514,11514,815,11514,815,11514,0,0,0',,,,,

Time Information       [HTDR:  1] Decoded below
  Transaction Timestamp          [sf : 1] 1411147899 (19-Sep-2014
17:31:39 GMT)
  Request Latency (ms)           [sf : 2] 221
  Response Latency (ms)          [sf : 3] 354
Subscriber             [HTDR:  2] Decoded below
  IMSI                           [sf : 1] 111113076999899
  IMEI                           [sf : 2] 999930769998990
  MSISDN                         [sf : 3] 3076999899
  SessionID                      [sf : 4] 2148630528
  Serving NW MCC                 [sf : 5] 226
  Serving NW MNC                 [sf : 6] 10
  Home PLMNID Length             [sf : 7] 5
  RAT Type                       [sf : 8] 6 (EUTRAN)
  Roaming Status                 [sf : 9] 1 (True)
  SGW/SGSN IP Address            [sf :10]
2606ae00b8001b000000000000000001 (2606:ae00:b800:1b00::1)
  APN (Mapped)                   [sf :11]
  APN (Received)                 [sf :12] reseller
Client Connection      [HTDR:  3] Decoded below
  Gi Network Context             [sf : 1] MVNO1
```

```
  Source IP Address                [sf : 2] 0a000003 (10.0.0.3)
  Source Port                      [sf : 3] 1024
  Destination IP Address           [sf : 4] 4a7de048 (74.125.224.72)
  Destination Port                 [sf : 5] 80
  IP-Transparency                  [sf : 6] 0 (Disabled)
HTTP Request           [HTDR:  4] Decoded below
  HTTP Request Method              [sf : 1] 2 (GET)
  Client Request URL (Orig)        [sf : 2]
http%3a%2f%2fwww.google.com%2f%3fgws_rd%3dssl
  Server Request URL (Real)        [sf : 3]
  Client Protocol                  [sf : 4] 0 (TCP)
  Server Protocol                  [sf : 5] 0 (TCP)
  User-Agent                      [sf : 6] Mozilla/4.0 (compatible; MSIE
6.0; Windows NT 5.1; .NET CLR 1.1.4322)
  Content-Type                     [sf : 7]
  Content-Length                   [sf : 8]
  Client Request Header Size       [sf : 9] 203
  Client Request Body Size         [sf :10] 0
  Server Request Header Size       [sf :11] 203
  Server Request Body Size         [sf :12] 0
  Range Request                    [sf :13] 0 (False)
  Request Successful               [sf :14] 1 (True)
  Chunked Request                  [sf :15] 0 (False)
  Explicit Proxy Request           [sf :16] 0 (False)
  DNS Resolution Applied           [sf :17] 0 (False)
  Proxy local IP Address           [sf :18] 9ba57302 (155.165.115.2)
  Proxy local IP port              [sf :19] 6279
  Origin Server IP                 [sf :20] 4a7de048 (74.125.224.72)
  Origin Server Port               [sf :21] 80
  Transaction Utilized BEF         [sf :22] 0 (Disabled)
HTTP Response          [HTDR:  5] Decoded below
  HTTP Response Status             [sf : 1] 200
  Server Content-Type              [sf : 2] text/html; charset=UTF-8
  Client Content-Type              [sf : 3] text/html; charset=UTF-8
  Server Content-Length            [sf : 4] 11,514
  Client Content-Length            [sf : 5] 11,514
  Server Response Header Size      [sf : 6] 815
  Server Response Body Size        [sf : 7] 11,514
  Client Response Header Size      [sf : 8] 815
  Client Response Body Size        [sf : 9] 11,514
  Chunked Response                 [sf :10] 0 (False)
  Served From Cache                [sf :11] 0 (False)
  Response Cached                  [sf :12] 0 (False)
Compression            [HTDR:  6] Field empty
Web Image Optimization [HTDR:  7] Field empty
Video Optimization     [HTDR:  8] Field empty
Minification           [HTDR:  9] Field empty
DNS Preemption         [HTDR: 10] Field empty
```

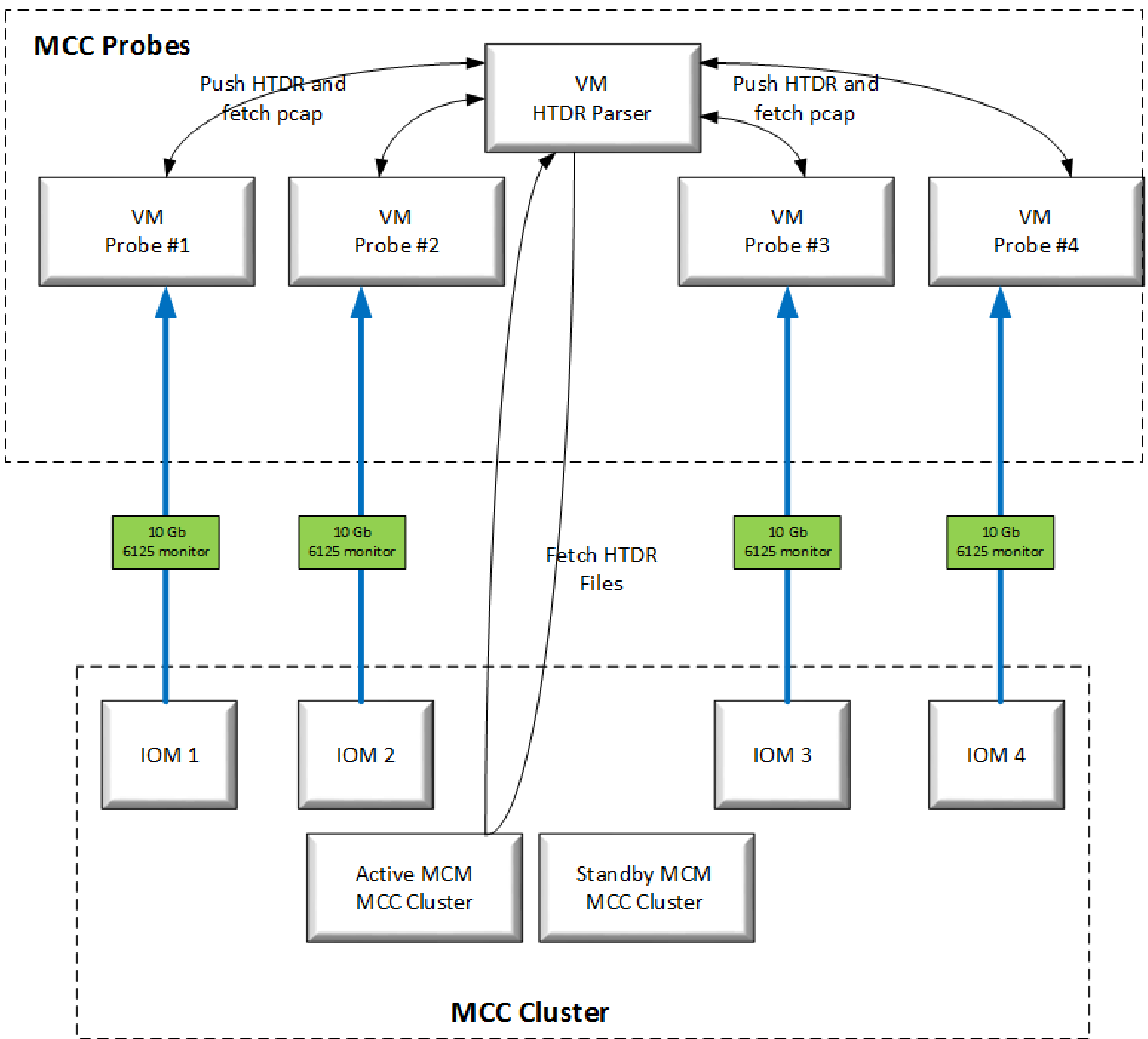
In the example highlighted text, HTDRs show subscriber information such as IMSI, IMEI, and MSISDN, the original private pre-NAT UE IP source address and port, and the public post-NAT UE IP source address and port.

1.13.1 HTDR Parser

HTTP Transaction Data Records (HTDR) can be used to parse through SGi pcaps to correlate SGi to a specific IMSI. Since most SGi traffic is proxy traffic, there is a public IP address on the MCC. The HTDR parser was developed to generate a SGi pcap for a specified IMSI.

Figure 1-1 shows the five VMs in the HTDR system. Four VMs are used for probes and one VM handles the HTDR files and correlation with the pcaps from the probes.

Figure 1-1 VMs in the HTDR System



1.13.2 HTDR Files

- All script files are stored in */home/root*.
- All HTDRs and pcaps are stored in */mnt/ramdisk*.

The following files are stored in the */home/root* directory:

- **htdr-parser.sh** – This is the main script that starts up during bootup. A crontab runs this script when the VM is booted. The scripts builds the ramdisk, starts ntp timing, and changes the hostname for the VM.
- **htdrfind.pl** – This is the main perl script that runs when the user wants to generate an SGi pcap. This calls the following scripts:
  - **htdrlist.sh** – Provides the directory output of the */mnt/ramdisk/htdr/htdrfiles* directory. It lists all current HTDR files.
  - **htdrget.sh** – Clears out all old HTDRs in the */mnt/ramdisk/htdr/htdrfiles/*. This calls the following script:
  - **getfile.sh** – Greps all current HTDRs on the MCC.
  - **htdrunziplist.sh** – Provides the directory output of the */mnt/ramdisk/htdr/unzip* directory. It lists all current HTDR files that are unzipped.
  - **puthtdrx.sh** – Sends the parsed HTDR file to each probe using SFTP.

1.13.2.1 Probe Files

The four probe VMs also have scripts in the */home/root* directory:

- **sdprobex.sh** – This is the main script that starts up during bootup. It starts the tshark that generates the pcaps. This script also creates the ramdisk and starts the ntp process. The following scripts are run from this script:
  - **htdrdirlist.sh** and **dirlistperl.pl** – Monitors the */mnt/ramdisk/zhtdr* directory to see if any new HTDR files have been sent to the probes. If a new HTDR file is seen, it parses the file with the following script:
  - **htdr_parser.pl** – Parses the HTDR and creates the pcap in the */mnt/ramdisk/zparsed-pcap/* directory.

1.13.2.2 Running the HTDR Parser

To run the HTDR parser, log into the Unix shell of the htdr-parser VM using Putty or any SSH client. The SSH port is 2222.

1. Use the following command to run the script:

```
root@HTDR-PARSER:~# perl /home/root/htdrfinder.pl
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = "en_US:en",
        LC_ALL = (unset),
```

```
        LANG = "en_US.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to the standard locale ("C").

Enter IMSI ------>
```

2 Enter the appropriate IMSI at the prompt.

If the IMSI is not present in any of the HTDRs, the following message is displayed:

```
sftp> IMSI 1234 is not present on any HTDR in the cluster
```

If the IMSI is present, the following output is displayed:

```
sftp>

 Files pushed waiting for output pcaps

my imsi is 111118352600990
root@HTDR-PARSER:~#
```

1.13.2.3 HTDR Output

The HTDR output may take between 1 and 20 minutes to parse, depending on the size of the SGi data. When a pcap is parsed, it is saved in the following directory on the HTDR VM: */mnt/ramdisk/zparsed-pcap/*

When all four probes have completed their parsing, the four following files should be present:

/mnt/ramdisk/raw-pcap/output1.pcap

/mnt/ramdisk/raw-pcap/output2.pcap

/mnt/ramdisk/raw-pcap/output3.pcap

/mnt/ramdisk/raw-pcap/output4.pcap

Run the following script on the HTDR VM to merge the four files into one file:

```
root@HTDR-PARSER:~# sh /home/root/mergefile.sh
```

The script creates the following merged pcap:

/mnt/ramdisk/parsed-pcap/merged_SGi.pcap

1.14 Miscellaneous CLI Commands

1.14.1 alarms active

Use the **`show alarms active`** command to display current alarm details.

```
an3000-mcm-slot7cpu1# show alarms active

SEQ ID  TRAP NAME       XPATH                     SNMPSEV  DATE TIME                     DETAILS
------------------------------------------------------------------------------------------------------
4484    anNtpSyncError  /infrastructure/ntpserver minor    2019-6-24 12:55:42.279 -0400  Cannot sync with
                                                                                         configured NTP servers.
```

See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for a list of alarms supported on the system.

1.14.2 clear-config

Use the **`clear-config`** command to remove all the configuration from the MCC.

```
an3000-mcm-slot7cpu1# clear-config
```

CAUTION: You must **never** run this command under any circumstances in a production environment.

1.14.3 commit

Use the **`commit`** command (in Config mode) to execute the current set of changes to the MCC.

```
an3000-mcm-slot7cpu1(config)# commit
```

1.14.4 configuration commit list

Use the **`show configuration commit list`** command to display a list of the most recent changes made to the MCC.

```
an3000-mcm-slot7cpu1# show configuration commit list
2019-06-20 17:23:48
SNo.  ID     User      Client    Time Stamp           Label  Comment
~~~~  ~~     ~~~~      ~~~~~~    ~~~~~~~~~~           ~~~~~  ~~~~~~~
0     10157  emsadmin  netconf   2019-06-17 07:13:33
1     10156  emsadmin  netconf   2019-06-12 15:23:11
2     10155  system    system    2019-06-12 13:58:11
3     10154  system    system    2019-06-12 13:58:10
4     10153  system    system    2019-06-12 13:58:10
```

1.14.5 revert

Use the **revert** command (in Config mode) to abandon all uncommitted changes in the MCC.

```
an3000-mcm-slot7cpu1(config)# revert
```

1.14.6 set-prompt

Use the **set-prompt** command to customize the CLI prompt.

The following special options are supported:

Note: *The '\' character is only supported for the options listed below.*

```
an3000-mcm-slot7cpu1# set-prompt
special options for prompts:
\d the date in "YYYY-MM-DD" format (e.g., "1978-06-10")
\h the hostname up to the first '.'
\H the hostname
\t the current time in 24-hour HH:MM:SS format
\T the current time in 12-hour HH:MM:SS format
\@ the current time in 12-hour am/pm format
\A the current time in 24-hour HH:MM format
\u the username of the current user
```

1.14.7 top

Use the **top** command to return to the top level of the CLI.

```
an3000-mcm-slot7cpu1# top
```

1.14.8 who

Use the **who** command to display currently logged on users. An asterisk appears next to the user who executed the command.

```
an3000-mcm-slot7cpu1# who
Session User     Context From          Proto Date     Mode
*31003  admin    cli     169.254.20.71 ssh   15:17:03 operational
 30369  emsadmin netconf 10.32.3.149   ssh   08:17:00 operational
```

Chapter 2

2 Session Management and Debugging

This section describes the recommended troubleshooting for problems related to EPS bearer activation and bearer traffic on the GGSN/PGW.

When you troubleshoot any issue, including Session Management, Affirmed Networks recommends that you check the active alarms and the cluster status using the following commands:

```
an3000-mcm-slot7cpu1# show cluster summary
an3000-mcm-slot7cpu1# show alarms active
an3000-mcm-slot7cpu1# show alarms history
an3000-mcm-slot7cpu1# show cluster node fault-summary
```

2.1 PDN Session Establishment Issues

Bearer activation issues occur for the following reasons:

- Signaling messages are not reaching the GGSN.
- Signaling messages are explicitly being rejected by the SGW or PGW.
- Signaling messages are accepted and the session is open but no data flows are created for the session.

2.1.1 Detecting Interface Connectivity Issues

Use the following command to determine whether session requests are being received by the MCC:

```
an3000-mcm-slot7cpu1# show zone default gateway debug-stats session-setup pdngw service-level | select num-session-setup-attempts
```

If session requests are being received it is assumed that the num-session-setup-attempts should be increasing when the command is run at 10-second intervals. If the session requests value is increasing, proceed to Step 1, otherwise go to Troubleshooting Gateway Signaling Rejects (page 2-2).

1 If requests are not being received, check network connectivity between the MCC and the adjacent GSN node (SGSN for 2G and 3G, MME for LTE) over the relevant interface. The ping test should always originate from the MCC's local endpoint. The following example shows commands for LTE troubleshooting:

```
an3000-mcm-slot7cpu1# ping <networkContext> -I <GTP_C_IP_Address>
<GSN_Peer_IP_address>
```

LTE example:

```
an3000-mcm-slot7cpu1# ping [networkContext] [MME IP] [-I MCC-
s11_loopback]

an3000-mcm-slot7cpu1# ping GnGp-NC 172.25.17.26 -I 107.77.171.1

PING 172.25.17.26 (172.25.17.26) from 107.77.171.1 : 56(84) bytes of
data.
64 bytes from 172.25.17.26: icmp_req=1 ttl=57 time=0.502 ms
64 bytes from 172.25.17.26: icmp_req=2 ttl=57 time=0.476 ms
```

If the ping test from the loopback to the MME is successful, the problem is likely to exist between the MME and UE. If the ping fails, proceed to Step 2.

2 If the ping fails, test connectivity to the gateway. For example:

```
an3000-mcm-slot7cpu1# ping GnGp-NC 172.20.128.41 -I 107.77.171.1
PING 172.20.128.41 (172.20.128.41) from 107.77.171.1 : 56(84) bytes of
data.
64 bytes from 172.20.128.41: icmp_req=1 ttl=64 time=0.455 ms
64 bytes from 172.20.128.41: icmp_req=2 ttl=64 time=0.424 ms
64 bytes from 172.20.128.41: icmp_req=3 ttl=64 time=0.447 ms
64 bytes from 172.20.128.41: icmp_req=4 ttl=64 time=0.441 ms
```

If the ping to the gateway from the loopback fails, a network connectivity issue exists between the MCC and the network.

If no connectivity issues are identified at this stage, verify the proper operation of the elements between the network and UE inclusive.

2.1.2 Troubleshooting Gateway Signaling Rejects

The MCC hosts GGSN/PGW functions, and in the absence of networking issues, see Detecting Interface Connectivity Issues (page 2-1), use these functions to investigate session establishment issues.

2.1.2.1 PGW Signaling Rejects

To detect signaling rejects related to the PGW, first determine the number of active PGW sessions/bearers.

Use the following command and check the value for the following parameters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance pgw
```

```
current-sessions
active-sessions                  Current number of active sessions
current-dual-stack-pdn-sessions  Current number of Dual Stack PDN
                                 Sessions
current-ipv4-pdn-sessions        Current number of IPv4 PDN Sessions
current-ipv6-pdn-sessions        Current number of IPv6 PDN Sessions
current-unique-subscribers       Current number of unique subscribers
bearers                          Current number of bearers
dedicated-bearers                Current number of dedicated bearers
default-bearers                  Current number of default bearers
```

Use the following command to determine which signaling procedure to the PGW failed:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance pgw | include rejects
```

```
create-session-rejects
create-session-rejects-malformed-requests
create-session-rejects-unknown-apn
create-session-rejects-resource-unavailable
create-session-rejects-service-not-supported
create-session-rejects-system-failure
create-session-rejects-misc
delete-session-rejects
ue-initiated-dedicated-bearer-activation-rejects
pgw-initiated-dedicated-bearer-activation-rejects
mme-initiated-dedicated-bearer-deactivation-rejects
ue-initiated-dedicated-bearer-deactivation-rejects
pgw-initiated-dedicated-bearer-deactivation-rejects
modify-bearer-command-rejects
update-bearer-rejects
modify-bearer-rejects
create-pdp-context-rejects
create-pdp-context-rejects-malformed-requests
create-pdp-context-rejects-unknown-apn
create-pdp-context-rejects-resource-unavailable
create-pdp-context-rejects-service-not-supported
create-pdp-context-rejects-system-failure
```

```
create-pdp-context-rejects-misc
delete-pdp-context-rejects
ue-initiated-secondary-pdp-activation-rejects
ggsn-initiated-secondary-pdp-activation-rejects
sgsn-initiated-secondary-pdp-deactivation-rejects
ue-initiated-secondary-pdp-deactivation-rejects
ggsn-initiated-secondary-pdp-deactivation-rejects
sgsn-initiated-update-pdp-rejects
ggsn-initiated-update-pdp-rejects
change-notification-rejects
change-notification-rejects-malformed-requests
change-notification-rejects-unknown-apn
change-notification-rejects-resource-unavailable
change-notification-rejects-service-not-supported
change-notification-rejects-system-failure
change-notification-rejects-misc
num-suspend-notification-rejects
num-resume-notification-rejects
create-session-rejects
```

You can determine the exact cause code by checking the debug log or the Wireshark trace configured for the subscriber.

2.1.2.2 SGW Signaling Rejects

To detect signaling rejects related to the SGW, first determine the number of active subscribers.

Use the following command and check the value for the following parameters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance sgw
```

```
active-pdn-connections        Current number of active PDN connections
active-sessions               Current number of active sessions
```

To determine the number of requests rejected by the SGW, check the value for the following parameters:

```
create-session-rejects-invalid-request
create-session-rejects-misc
create-session-rejects-system-failure
create-bearer-mme-rejects
create-bearer-rejects
create-bearer-rejects-invalid-requests
create-bearer-rejects-misc
create-bearer-rejects-system-failure
bearer-resource-fail-indication-invalid-requests
```

```
bearer-resource-fail-indication-misc
bearer-resource-fail-indication-system-failure
```

You can determine the exact cause code by checking the debug log or the Wireshark trace configured for the subscriber.

2.2 PDN Session Payload Issues

When the bearer activation is successful but there is no payload or the expected payload values are not met, a payload handling issue exists.

The problem can be divided into two basic categories:

- Payload does not reach the GGSN/PGW
- Payload is discarded by the GGSN/PGW

1 Check the uplink and downlink counters for packet drops.

Use the following commands to verify that uplink packets are reaching the gateway:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance pgw GGSN | include bytes
```

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance pgw GGSN | include packets
```

Check the uplink counters:

```
uplink-bytes
uplink-packets
```

If the uplink counters are not increasing, it means that traffic is not being received by the gateway and you need to troubleshoot the network connectivity. To determine if downlink packets are being sent by the SAEGW, check the following counters:

```
downlink-bytes
downlink-packets
```

2 Use the following command to get more detailed information and check if the packets are being received and sent for a specific subscriber:

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <id>
```

Check the following counters:

```
uplink-green-packets
uplink-green-bytes
downlink-green-packets
```

The same output gives detailed counters about the condition of dropped packets:

```
uplink-service-drop-packets
uplink-service-drop-bytes
uplink-no-tft-packets
uplink-no-route-packets
```

```
uplink-ttl-exhaust-packets
uplink-bad-header-packets
uplink-bad-ip-addr-packets
uplink-mtu-discard-packets
uplink-transmit-icmp-packets
uplink-send-error-packets
uplink-send-error-bytes
downlink-tcp-retransmission-packets
downlink-tcp-retransmission-bytes
downlink-diverted-returned-packets
downlink-error-drop-packets
downlink-error-drop-bytes
downlink-service-drop-packets
downlink-service-drop-bytes
downlink-no-tft-packets
downlink-no-route-packets
downlink-ttl-exhaust-packets
downlink-mtu-discard-packets
downlink-transmit-icmp-packets
downlink-bad-header-packets
downlink-send-error-packets
downlink-send-error-bytes
uplink-hairpin-policy-discard-packets
uplink-hairpin-error-discard-packets
```

2.3 Gx Issues

During session establishment and maintenance, the PCEF also communicates with the PCRF.

Use the following command to check for Diameter peering issues:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-
group diameter peer-status
service-construct interface aaa-interface-group diameter peer-status
HOST
NODE NAME  HOST NAME                     REALM              IP ADDRESS     PORT  TRANSPORT  ROLE     NAME  STATUS
-------------------------------------------------------------------------------------------------------------------
OCS        an3000p.ocs.affirmed.local   affirmed.ocsserver  10.224.205.73  3868  tcp        primary        ok
```

You should see the status of the peer as **ok**.

If the status is inactive, check the following output:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-group diameter peer-stats
service-construct interface aaa-interface-group diameter peer-stats
                                                  NUM       NUM       NUM
                                                  PROTOCOL  PERMANENT ROUTING  NUM OUT  NUM OUT   NUM IN   NUM IN
NODE NAME  HOST NAME     REALM                   ERROR     FAILURE   FAILURE  REQUEST  RESPONSE  REQUEST  RESPONSE
-----------------------------------------------------------------------------------------------------------------------
OCS        an3000p.local aff.ocsserver           0         0         0        3938     6         6        3930
```

Protocol errors occur at the base protocol level and may require per-hop attention. For example, a message routing error is a protocol error.

Common Diameter protocol errors are:

(3002)DIAMETER_UNABLE_TO_DELIVER (Initial/Terminate)The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.

(3004) DIAMETER_TOO_BUSY(Initial/Update/Terminate)The diameter device cannot deliver the message to the desired destination.

(3005) DIAMETER_LOOP_DETECTED (Initial/Terminate) A diameter agent detected a loop while trying to get the message to the desired destination.

Permanent failure errors (5xxx) inform the peer that the request failed and should not be re-attempted.

Use the following parameters to configure whether or not to send a terminate message upon receipt of a permanent failure error message:

```
service-construct interface pcrf-charging PAS_INTF
pcrf-cca-error-send-terminate true
```

Check the response counters to determine whether the PCRF server is replying to the PCEF. The PCRF server is replying to the messages when the responses match the requests. If the server is replying, check the Wireshark trace for a Diameter Success (Cause Code 2001) for the CEA.

If there is a Diameter success message from the PCEF, check the gateway logs. The log should contain the reason for not establishing the peer connection (e.g. CEA timeout, realm-name/host-name mismatch).

Use the following command to check the CEA timeout value and the hostname/realmname on the gateway side:

```
an3000-mcm-slot7cpu1# show running-config service-construct interface aaa-interface-group diameter node-admin <pcrf-interface-name>
```

Use the following command if the Diameter peer is established but you still observe a large number of CCR/CCA failures:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-group diameter statistics cluster-level peer interface-type GX <pcrf-interface-name>
```

2.3.1 Gx Failure Handling

If the Diameter link between the PCEF and PCRF fails, you can configure the MCC to continue to set up the call using a locally configured policy.

Use the following commands to configure the call setup:

```
an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-
charging PAS_INTF
```

```
an3000-mcm-slot7cpu1(config)# pcrf-down-default-gx-call-setup enabled
```

Though service may be restored, any network functionality supported by the Gx interface may not be available (for example, charging, blocking for insufficient balance, throttling).

2.3.2 Default QoS Policy

If the Gx service fails on the MCC and the pcrf-down-default-gx-call-setup flag is set to enabled, the default behavior of the MCC is to assign the user with a default QoS policy. You can use counters that track the assignment of the default QoS policy to diagnose Gx issues. If the counters are increasing, assume there is an underlying issue.

Use the following command to display the relevant counters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-
level call-performance pgw | include default
```

```
default-bearers                                                  37950
interval-peak-default-bearers                                    39385
number-of-sessions-setup-with-default-policy                       735
number-of-active-sessions-with-default-policy                        2
delete-pdp-context-drops-teardown-ind-missing-default-bearer         0
```

2.4 CALEA Issues

The Communications Assistance for Law Enforcement Act (CALEA) interface provides surveillance capabilities to assist law enforcement. For details, see the *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide*.

To execute any CALEA CLI commands, you must be logged in as the CALEA user:

- Username: calea
- Default Password: calea

2.4.1 show calea interface-status

Use the **show calea interface-status** command to display the status and utilization of the individual binds between the relevant CALEA interface and the remote server. To display the CALEA Interface Table status, you must be in Exec mode.

```
an3000-mcm-slot7cpu1# show calea interface-status
calea interface-status x2-ber-hi-allen
 transport-type                                   tcp
 interface-type                                   x2
 interface-format-type                            HI_format
 operational-status                               up
 number-of-agent-instances                        1
 number-of-tpkt-keep-alive-messages-sent          11192
 number-of-tpkt-keep-alive-ber-encode-fail-count 0
 number-of-tpkt-keep-alive-xer-encode-fail-count 0
calea interface-status x2-ber-hi-bothell
 transport-type                                   tcp
 interface-type                                   x2
 interface-format-type                            HI_format
 operational-status                               up
 number-of-agent-instances                        1
 number-of-tpkt-keep-alive-messages-sent          11203
 number-of-tpkt-keep-alive-ber-encode-fail-count 0
 number-of-tpkt-keep-alive-xer-encode-fail-count 0
calea interface-status x3-ber-hi-allen
 transport-type                                   tcp
 interface-type                                   x3
 interface-format-type                            HI_format
 operational-status                               up
 number-of-agent-instances                        1
 number-of-tpkt-keep-alive-messages-sent          11172
 number-of-tpkt-keep-alive-ber-encode-fail-count 0
 number-of-tpkt-keep-alive-xer-encode-fail-count 0
calea interface-status x3-ber-hi-bothell
 transport-type                                   tcp
 interface-type                                   x3
 interface-format-type                            HI_format
 operational-status                               up
 number-of-agent-instances                        1
 number-of-tpkt-keep-alive-messages-sent          11203
 number-of-tpkt-keep-alive-ber-encode-fail-count 0
 number-of-tpkt-keep-alive-xer-encode-fail-count 0
```

2.4.2 show calea interface-group-status

Use the **interface-group-status** parameter to display details on configuration and performance for the CALEA group defined on the MCC. To display the CALEA Interface Group status, you must be in Exec mode.

```
an3000-mcm-slot7cpu1# show calea interface-group-status
calea interface-group-status x2-ber-hi-allen
 role                                                          primary
 interface-group-type                                          x2
 output-type                                                   ber
 interfaceFormat-type                                        HI_format
 operational-status                                            up
 iri-buffer-size                                               1024000
 iri-buffer-current-size                                       0
 iri-buffer-num-packets                                        0
 iri-buffer-high-water-mark                                    0
 iri-buffer-num-buffer-size-exceeded                           0
 iri-buffer-num-dropped-packets                                0
 number-of-agent-instances                                     1
 number-of-x2-events-offered                                   21
 number-of-transmit-ok                                         21
 number-of-transmit-fail-bad-interface-group                   0
 number-of-transmit-fail-interface-group-down                  0
 number-of-transmit-fail-no-space                              0
 number-of-x2-encode-fail-no-buffer                            0
 number-of-x2-ber-encode-fail                                  0
 number-of-x2-xer-encode-fail                                  0
 number-of-targets                                             0
 total-cumulative-LI-Sessions                                  10
 currently-active-li-calls                                     0
 total-number-of-intercepted-packets-for-current-active-calls  0
 number-of-x2-events-for-current-active-calls                  0
 total-number-of-all-terminated-LI-sessions                    10
 total-number-of-terminated-LI-sessions-due-to-call-disconnect 8
 total-number-of-terminated-LI-sessions-due-to-deprovision     2
calea interface-group-status x2-ber-hi-bothell
 role                                         secondary
 interface-group-type                         x2
 output-type                                  ber
 interfaceFormat-type                         HI_format
 operational-status                           up
 number-of-agent-instances                    1
 number-of-transmit-ok                        0
 number-of-transmit-fail-bad-interface-group  0
 number-of-transmit-fail-interface-group-down 0
 number-of-transmit-fail-no-space             0
calea interface-group-status x3-ber-hi-allen
 role                                                        primary
```

```
 interface-group-type                                       x3
 output-type                                                ber
 interfaceFormat-type                                      HI_format
 operational-status                                         up
 cc-buffer-size                                             51200000
 cc-buffer-current-size                                     0
 cc-buffer-num-packets                                      0
 cc-buffer-high-water-mark                                  0
 cc-buffer-num-buffer-size-exceeded                         0
 cc-buffer-num-dropped-packets                              0
 number-of-agent-instances                                  1
 number-of-x3-events-offered                                48970
 number-of-x3-event-encode-failures-no-buffer               0
 number-of-x3-event-xer-encode-failures                     0
 number-of-x3-event-ber-encode-failures                     0
 number-of-transmit-ok                                      48970
 number-of-transmit-fail-bad-interface-group                0
 number-of-transmit-fail-interface-group-down               0
 number-of-transmit-fail-no-space                           0
 number-of-x3-encode-fail-no-buffer                         0
 number-of-x3-ber-encode-fail                               0
 number-of-x3-xer-encode-fail                               0
 number-of-targets                                          0
 total-cumulative-LI-Sessions                               6
 currently-active-li-calls                                  0
 total-number-of-intercepted-packets-for-current-active-calls 0
 number-of-x3-events-for-current-active-calls               0
calea interface-group-status x3-ber-hi-bothell
 role                                          secondary
 interface-group-type                          x3
 output-type                                   ber
 interfaceFormat-type                          HI_format
 operational-status                            up
 number-of-agent-instances                     1
 number-of-transmit-ok                         0
 number-of-transmit-fail-bad-interface-group   0
 number-of-transmit-fail-interface-group-down  0
 number-of-transmit-fail-no-space              0
```

2.4.3 show calea service-status

Use the **`service-status`** parameter to display a consolidated view of CALEA across all interfaces (X1, X2, X3). To display the CALEA service status, you must be in Exec mode.

```
an3000-mcm-slot7cpu1# show calea service-status
calea service-status all
 operational-status                                              up
 number-of-agent-instances                                       1
```

```
number-of-x1-1-interfaces                                        0
number-of-x2-interfaces                                          2
number-of-x3-interfaces                                          2
number-of-bad-events-received                                    0
number-of-rx-events-dropped-no-target                            0
number-of-x2-events-received                                     21
number-of-x3-events-received                                    56821
number-of-transmit-ok                                           48991
number-of-transmit-fail-bad-interface                            0
number-of-transmit-fail-interface-down                           0
number-of-transmit-fail-no-space                                 0
number-of-echo-requests-received                                 0
number-of-echo-requests-sent                                     0
number-of-echo-responses-received                                0
number-of-echo-responses-sent                                    0
number-of-echo-timeouts                                          0
number-of-targets                                                0
total-cumulative-LI-Sessions                                     10
currently-active-li-calls                                        0
currently-camp-on-triggers                                       0
number-of-x2-output-events                                       21
number-of-x3-output-events                                      48970
number-of-x2-output-events-failed                                0
number-of-x2-events-dropped-no-interface                         0
number-of-x2-encode-fail-no-buffer                               0
number-of-x2-xer-encode-fail                                     0
number-of-x2-ber-encode-fail                                     0
number-of-x3-output-events-failed                                0
number-of-x3-events-dropped-no-interface                         0
number-of-x3-event-encode-failures-no-buffer                     0
number-of-x3-xer-encode-fail                                     0
number-of-x3-ber-encode-fail                                     0
total-number-of-intercepted-packets-for-current-active-calls     0
number-of-x2-events-for-current-active-calls                     0
number-of-x3-events-for-current-active-calls                     0
total-number-of-all-terminated-LI-sessions                       10
total-number-of-terminated-LI-sessions-due-to-call-disconnect    8
total-number-of-terminated-LI-sessions-due-to-deletion-of-context 0
total-number-of-terminated-LI-sessions-due-to-deprovision        2
```

2.5 GTPP Issues

GTP Prime (GTPP) is an IP-based protocol implemented over the Ga interface that can be used with both TCP and UDP for transporting charging data from the gateways (Charging Data Function (CDF)) to the Charging Gateway Function (CGF). The CDF function implemented in the MCC is bound to two CGFs in active/standby mode and the service is hosted in the SUPPORTZONE.

Before proceeding, check the active alarms and the cluster status as described in Session Management and Debugging (page 2-1).

2.5.1 gtpp node-status

Use the **show service-construct interface aaa-interface-group gtpp node-status** command to display the operational status of the local GTPP instance:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-
group gtpp node-status
service-construct interface aaa-interface-group gtpp node-status
                                                                                         SEQUENCE
                                      OPERATIONAL HOST    HOST IP       DROPPED  DUPLICATE  OVER RUN
NAME      ADMINSTATUS OPERSTATUS  REASON      PORT    ADDRESS       PACKETS  PACKETS    PACKETS
----------------------------------------------------------------------------------------------------
GA-XXXX   enabled     inservice   -           6101    107.77.163.9  0        0          0
```

2.5.2 gtpp peer-status

Use the **gtpp peer-status** parameter to check the GTPP peering state between the MCC and the GTPP endpoint:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-
group gtpp peer-status
service-construct interface aaa-interface-group gtpp peer-status GA-
XXXX NDC2C_OCG7_1
 ip-address                172.18.250.231
 port                      6101
 transport                 udp
 role                      primary
 adminstatus               enabled
 operstatus                okay
 dataRecordRetryCount      3
 dataRecordRespTimeout     5
 echoRetryCount            3
 echoRespTimer             10
 periodicHealthCheckTimer  60
service-construct interface aaa-interface-group gtpp peer-status GA-
XXXX NDC2C_OCG7_2
 ip-address                172.18.250.236
 port                      6201
 transport                 udp
 role                      secondary
 adminstatus               enabled
 operstatus                okay
 dataRecordRetryCount      3
 dataRecordRespTimeout     5
 echoRetryCount            3
 echoRespTimer             10
 periodicHealthCheckTimer  60
```

If the peer-status is enabled but the operational status is not `okay`, you need to verify connectivity between the MCC and the peer. Use the host IP address (see gtpp node-status (page 2-13) for the host IP address) as the source address and send the ping from the SUPPORTZONE.

```
an3000-mcm-slot7cpu1# ping ZZZZ 172.18.250.231 -I 107.77.163.9
Warning: Permanently added '[169.254.100.9]:2020' (RSA) to the list of
known hosts.
PING 172.18.250.231 (172.18.250.231) from 107.77.163.9 : 56(84) bytes
of data.
64 bytes from 172.18.250.231: icmp_req=1 ttl=252 time=0.405 ms
64 bytes from 172.18.250.231: icmp_req=2 ttl=252 time=0.322 ms
64 bytes from 172.18.250.231: icmp_req=3 ttl=252 time=0.427 ms
64 bytes from 172.18.250.231: icmp_req=4 ttl=252 time=0.348 ms
64 bytes from 172.18.250.231: icmp_req=5 ttl=252 time=0.351 ms
```

2.5.3 gtpp peer-stats

Use the **`gtpp peer-stats`** parameter to display a consolidated view of the GTPP performance giving the number of aggregate messages and generic errors sent to and from a peer.

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-
group gtpp peer-stats
service-construct interface aaa-interface-group gtpp peer-stats
                        NUM         NUM         NUM                                                 NUM
                        PROTOCOL    PERMANENT   ROUTING     NUM OUT     NUM OUT     NUM IN      NUM IN      LINK
HOST NAME   NAME        ERROR       FAILURE     FAILURE     REQUEST     RESPONSE    REQUEST     RESPONSE    BOUNCES
-------------------------------------------------------------------------------------------------------------------
GA-XXXX     YYYY_ZZZ_1  0           0           0           4229        73912       73912       3931        0
GA-XXXX     YYYY_ZZZ_2  0           0           0           2463        73913       73913       2463        0
```

2.5.4 gtpp statistics

Use the **`gtpp statistics`** parameter to display information about the status of communication between each GTPP peer.

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-interface-
group gtpp statistics
service-construct interface aaa-interface-group gtpp statistics
cluster-level peer node-name GA-XXXX YYYY_ZZZ_1
 num-echo-req-tx                             2462
 num-echo-req-rx                             73880
 num-echo-resp-tx                            73880
 num-echo-resp-rx                            2462
 num-node-alive-req-tx                       3
 num-node-alive-req-rx                       0
 num-node-alive-resp-tx                      0
 num-node-alive-resp-rx                      0
 num-redirection-req-rx                      0
 num-redirection-resp-tx                     0
```

```
 num-data-record-transfer-req-tx             1763
 num-data-record-transfer-resp-rx            1466
 num-potential-duplicate-data-record-rx      0
 num-potential-duplicate-empty-data-record-tx 0
 num-cancel-request-tx                       0
 num-release-request-tx                      0
 num-req-fulfilled-rx                        0
 num-req-accepted-rx                         0
 Discontinuity Timestamp                     1430802249

service-construct interface aaa-interface-group gtpp statistics
cluster-level peer node-name GA-XXXX YYYY_ZZZ_2
 num-echo-req-tx                             2462
 num-echo-req-rx                             73882
 num-echo-resp-tx                            73882
 num-echo-resp-rx                            2462
 num-node-alive-req-tx                       3
 num-node-alive-req-rx                       0
 num-node-alive-resp-tx                      0
 num-node-alive-resp-rx                      0
 num-redirection-req-rx                      0
 num-redirection-resp-tx                     0
 num-data-record-transfer-req-tx             0
 num-data-record-transfer-resp-rx            0
 num-potential-duplicate-data-record-rx      0
 num-potential-duplicate-empty-data-record-tx 0
 num-cancel-request-tx                       0
 num-release-request-tx                      0
 num-req-fulfilled-rx                        0
 num-req-accepted-rx                         0
 Discontinuity Timestamp                     1430802249
```

2.6 GiGW Subscriber Learning Limitations

The following behavior is related to GiGW session learning from RADIUS Accounting-Request messages.

- For emergency sessions (IMSI==0) without MSISDN (Calling-Station-Id), GiGW maintains only one session, therefore, if another emergency call occurs, MCC replaces the first session with the second session.
- For fixed-line subscribers, the username takes precedence over MSISDN. That is, if a username exists, MCC uses it to anchor the pseudo IMSI. If no username exists, MSISDN is used to anchor the pseudo IMSI.
- For WAG subscribers, the username and Calling-Station-Id should be in MAC address format.

2.7 SLB Load Status Table

The Steering Load Balancer (SLB) load status table displays the load distribution status across the gateways.

```
an3000-mcm-slot7cpu1# show zone default gateway slb-load-status
gateway slb-load-status
GATEWAY                                           LOAD          CURRENT    ABSOLUTE
TYPE      NAME                         SERVICEID  STATE         LOAD       VALUE
-------------------------------------------------------------------------------------
sgw       SGW_100_11_1_0_SGW_0         152        MinorLow      0          0
sgw       SGW_100_11_1_12_SGW_24       741        MinorLow      0          0
sgw       SGW_100_11_1_16_SGW_32       747        MinorLow      0          0
sgw       SGW_100_11_1_20_SGW_40       751        MinorLow      0          0
sgw       SGW_100_11_1_24_SGW_48       755        MinorLow      0          0
sgw       SGW_100_11_1_28_SGW_56       759        MinorLow      0          0
sgw       SGW_100_11_1_4_SGW_8         699        MinorLow      0          0
sgw       SGW_100_11_1_8_SGW_16        721        MinorLow      0          0
sgw       SGW_100_12_1_13_SGW_26       742        MinorLow      0          0
sgw       SGW_100_12_1_17_SGW_34       748        MinorLow      0          0
sgw       SGW_100_12_1_1_SGW_2         164        MinorLow      0          0
sgw       SGW_100_12_1_21_SGW_42       752        MinorLow      0          0
sgw       SGW_100_12_1_25_SGW_50       756        MinorLow      0          0
sgw       SGW_100_12_1_29_SGW_58       760        MinorLow      0          0
sgw       SGW_100_12_1_5_SGW_10        703        MinorLow      0          0
sgw       SGW_100_12_1_9_SGW_18        725        MinorLow      0          0
sgw       SGW_100_13_1_10_SGW_20       730        MinorLow      0          0
sgw       SGW_100_13_1_18_SGW_36       749        MinorLow      0          0
sgw       SGW_100_13_1_22_SGW_44       753        MinorLow      0          0
sgw       SGW_100_13_1_26_SGW_52       757        MinorLow      0          0
sgw       SGW_100_13_1_2_SGW_4         176        MinorLow      0          0
sgw       SGW_100_13_1_30_SGW_60       761        MinorLow      0          0
sgw       SGW_100_13_1_6_SGW_12        708        MinorLow      0          0
sgw       SGW_100_14_1_11_SGW_22       736        MinorLow      0          0
sgw       SGW_100_14_1_15_SGW_30       744        MinorLow      0          0
sgw       SGW_100_14_1_19_SGW_38       750        MinorLow      0          0
sgw       SGW_100_14_1_23_SGW_46       754        MinorLow      0          0
sgw       SGW_100_14_1_27_SGW_54       758        MinorLow      0          0
```

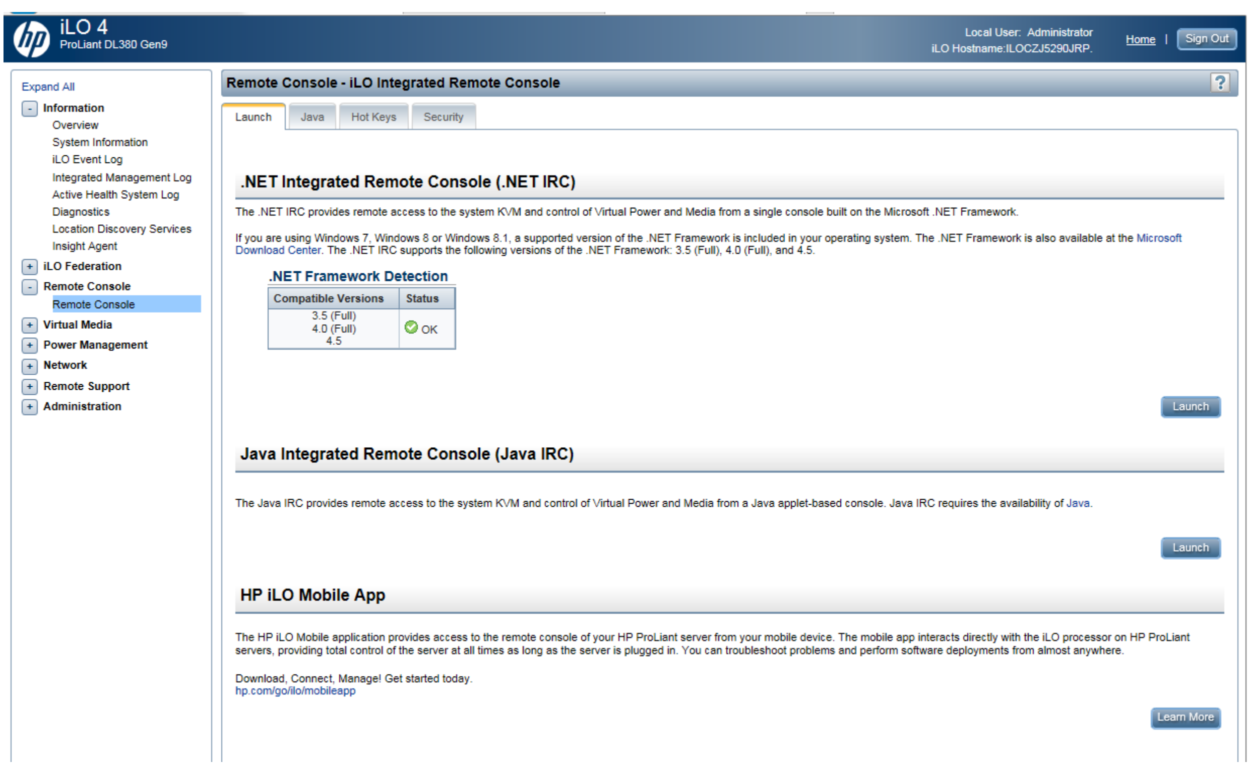
```
pgw        PDNGW_100_11_1_0_PDNGW_0     145         MinorLow      0          0
pgw        PDNGW_100_11_1_12_PDNGW_24   589         MinorLow      0          0
pgw        PDNGW_100_11_1_16_PDNGW_32   611         MinorLow      0          0
pgw        PDNGW_100_11_1_20_PDNGW_40   633         MinorLow      0          0
pgw        PDNGW_100_11_1_24_PDNGW_48   655         MinorLow      0          0
pgw        PDNGW_100_11_1_28_PDNGW_56   677         MinorLow      0          0
pgw        PDNGW_100_11_1_4_PDNGW_8     545         MinorLow      0          0
pgw        PDNGW_100_11_1_8_PDNGW_16    567         MinorLow      0          0
pgw        PDNGW_100_12_1_13_PDNGW_26   593         MinorLow      0          0
pgw        PDNGW_100_12_1_17_PDNGW_34   615         MinorLow      0          1
pgw        PDNGW_100_12_1_1_PDNGW_2     157         MinorLow      0          0
pgw        PDNGW_100_12_1_21_PDNGW_42   637         MinorLow      0          0
pgw        PDNGW_100_12_1_25_PDNGW_50   659         MinorLow      0          0
pgw        PDNGW_100_12_1_29_PDNGW_58   681         MinorLow      0          0
pgw        PDNGW_100_12_1_5_PDNGW_10    549         MinorLow      0          0
pgw        PDNGW_100_12_1_9_PDNGW_18    571         MinorLow      0          0
```

2.8 Accessing the Remote Console

Figure 2-1 shows the Remote Console screen. Use .NET Integrated Remote Console (.NET IRC) to access the blade.

JAVA IRC and the Mobile App are also valid access methods, but these have more dependencies, so .NET IRC is used in this example.

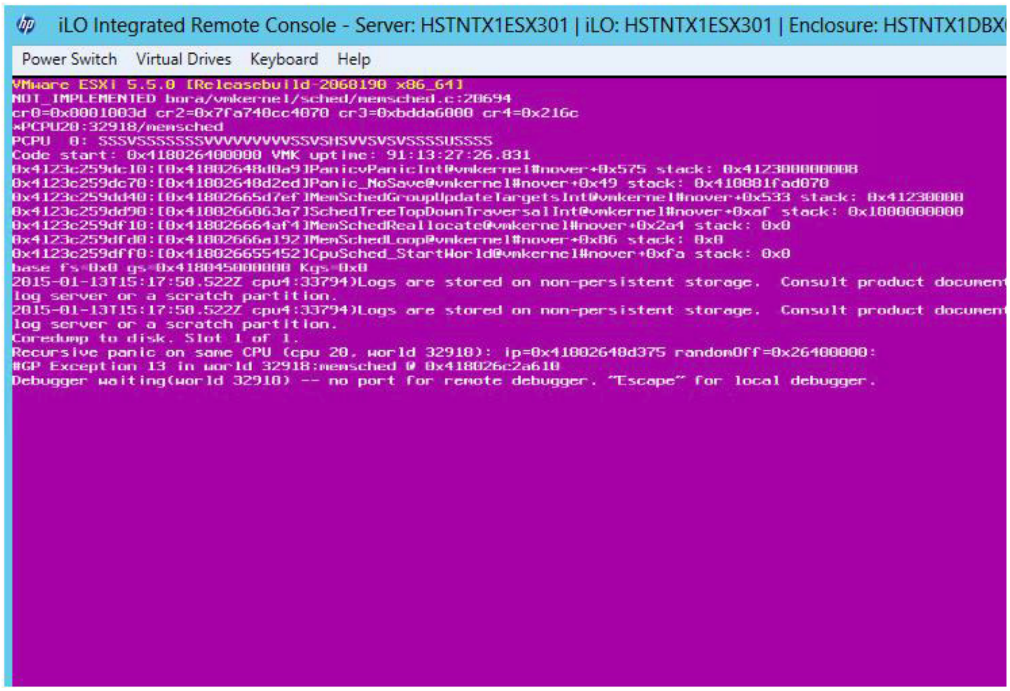
Figure 2-1 Remote Console Screen



2.8.1 Recovering from a Fatal Error

When the VM kernel experiences a fatal error, the system displays a purple diagnostic screen. This means the blade and all VMs hosted on the blade are down, as shown by the **`show cluster summary`** CLI command. All nodes hosted on the same blade will show the CPU state as down at the same time. Figure 2-2 shows the diagnostic screen.

Figure 2-2 Fatal Error Diagnostic Screen



Before rebooting the blade, collect the fault files that are stored in the */var/log/core* folder. The Affirmed TAC uses these files to diagnose issues.

Figure 2-3 shows the **Remote Console Power Switch** screen. Reboot the blade using one of the reboot options from the Power Switch menu. Table 2-1 describes the function of each option.

The blade normally takes approximately five minutes to reboot. Track the progress of the VMs using the `show cluster` CLI command.

Figure 2-3 Remote Console Power Switch

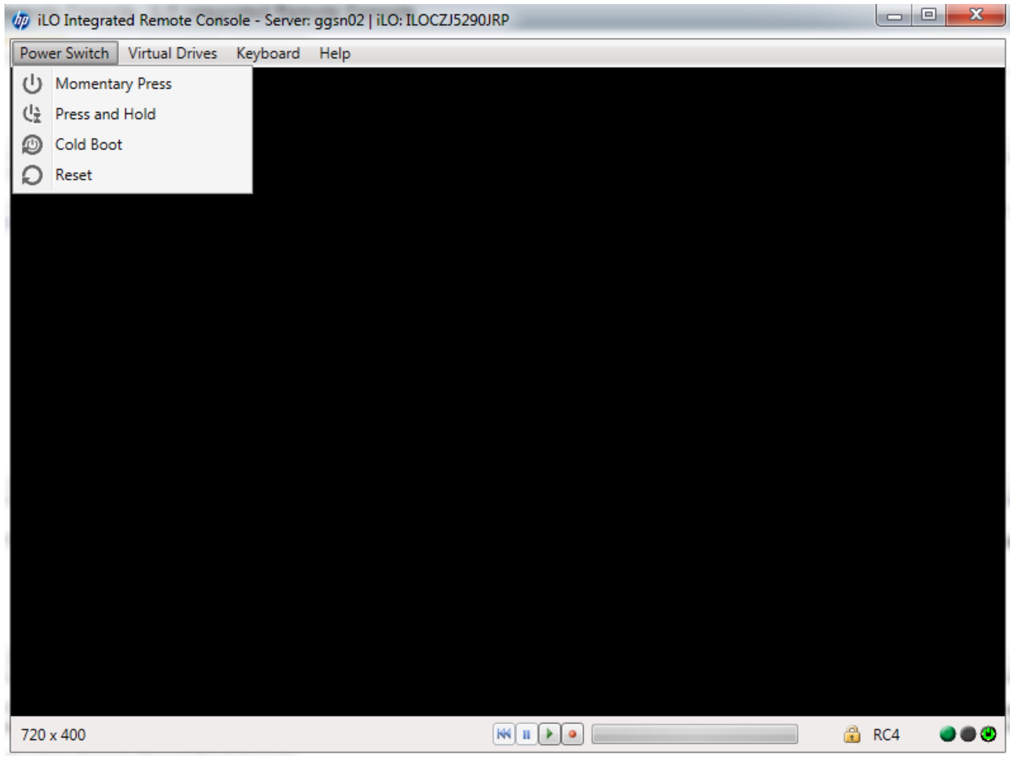


Table 2-1 Remote Console Power Switch Options

Options	Description
Momentary Press	The same as pressing the physical power button. If a server is powered off, a momentary press will turn the server power on.
Press and Hold	The same as pressing the physical power button for five seconds and then releasing it.
Cold Boot	Immediately removes power from the server. Processors, memory, and I/O resources lose main power. The server will restart after approximately six seconds. This option circumvents the graceful shutdown features of the operating system.
Reset	Forces the server to warm-boot. CPUs and I/O resources are reset. This option circumvents the graceful shutdown features of the operating system.

Chapter 3

3 Configuring Interfaces

This section describes useful information on configuring interfaces.

3.1 Configuring the SWu as an Access Interface

An access interface is any ip-interface that handles Encapsulated Security Payload (SWu interface to a UE) or L2oGRE (towards an access point) traffic. Any ip-interface handling that type of traffic *must* be configured as an access interface.

Use the following command to configure the SWu as an access interface:

```
an3000-mcm-slot7cpu1(config)# network-context <network-context-name>
access-interface <ip-interface>
```

where

Parameter	Description
`network-context`	Specifies the name of the access network context.
`access-interface`	Specifies the IP Interface to be designated as an access interface.

Chapter 4

4 Technical Support

This section describes:

- How to access the Affirmed Networks customer portal to open a support ticket
- What to include when creating a ticket
- Contacting the Technical Assistance Center (TAC)

4.1 Opening a Support Ticket

For non-critical cases, Affirmed Networks recommends that customers use the online ticket system as the primary option to report issues. The system allows Affirmed support to track a ticket during its lifetime (for SLA adherence), communicate with the customer, and escalate tickets to engineering. Access the customer portal at: https://system.netsuite.com/app/login/secure/privatelogin.nl?c=3505467

Figure 4-1 shows the customer portal home page. Click the **Contact Support** link to open a ticket.

Figure 4-1 Open a New Case in the Customer Portal

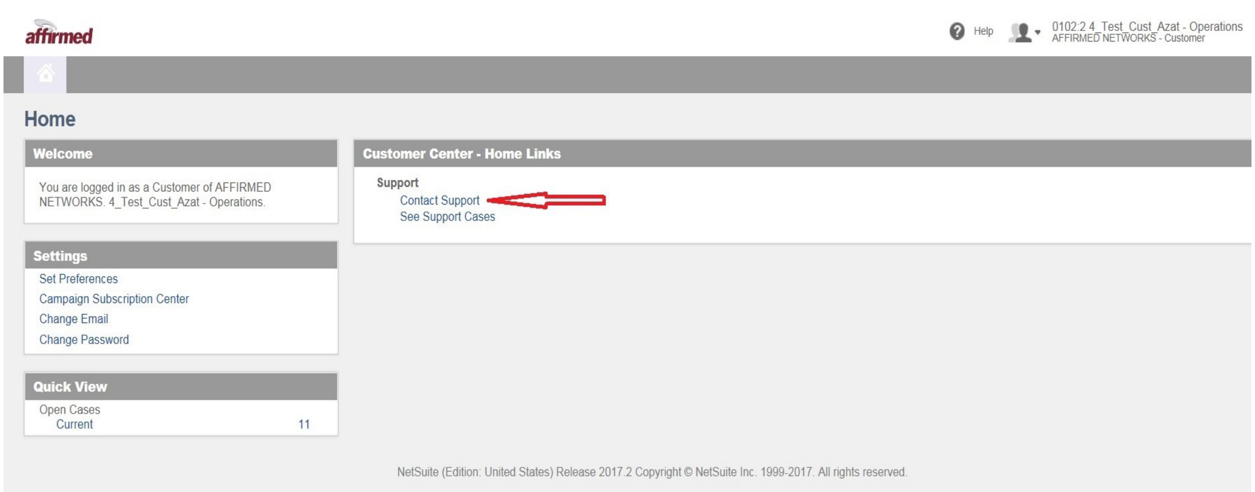
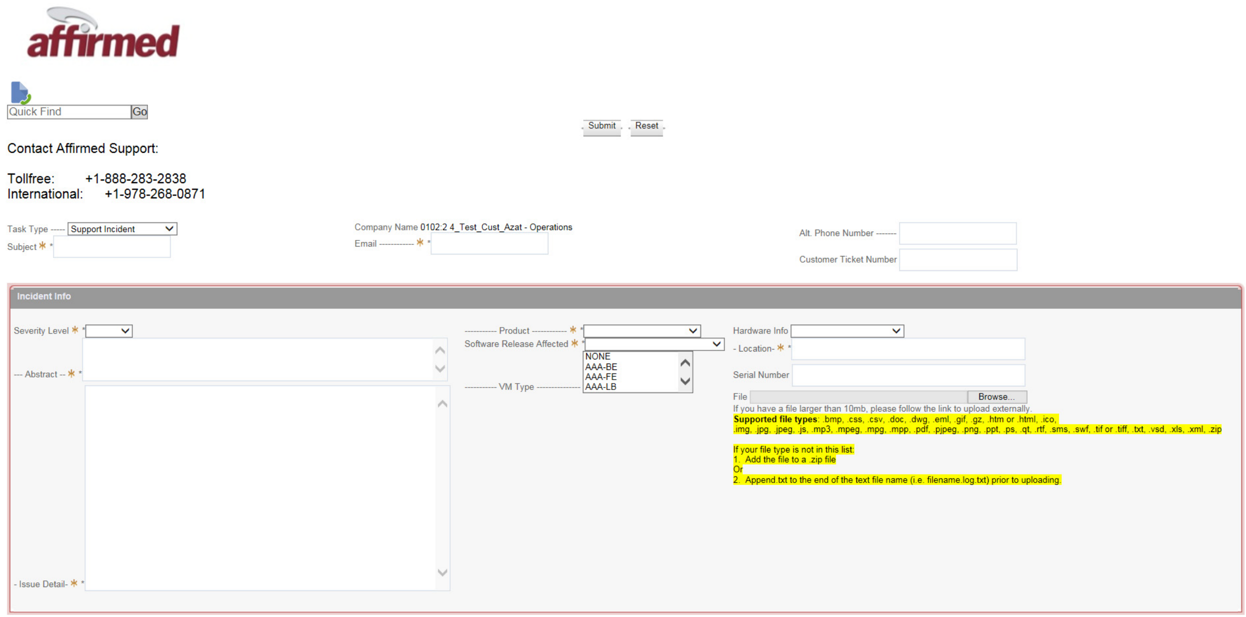


Figure 4-2 shows the required fields, marked with a ✱, to open a new case.

Figure 4-2 New Case Required Fields



4.1.1 Completing the Ticket Information in the Customer Portal

Complete the following fields to create a ticket:

1 **Incident Subject:**

Enter a brief description of the issue.

2 **Severity Level:**

Select the severity level for the issue from the drop-down list: Critical, Major, Minor, or Info.

3 **Product:**

Select the product. For example, PGG, EMS, and vCenter.

4 **Software Release Affected:**

Enter the software release that is affected.

Use the **`show image-management status`** command to obtain the release number:

```
image-management status active 18 7 1
 admin-version       9.4.0.0-392.REL
 operational-version 9.4.0.0-392.REL
 activation-status   Idle
 rollback-version    9.4.0.0-360.REL
 prepared-version    NotAvailable
```

5 **Location:**

Enter the location of the site where the issue exits.

6 **Issue Detail:**

Describe the issue in as much detail as possible and include the following:

- A percentage estimate of the number of affected subscribers.
- Services that are affected.
- Information derived from troubleshooting.
- Events that may have led to the issue.

7 **Technical Support Information Logs (when applicable):**

Use the **`save-tech-support-info`** command to create detailed system logs that the Affirmed TAC uses to diagnose issues. These logs can be very large; check that the file system has sufficient space before you generate the logs. The */public* folder should be below 80%.

```
an3000-mcm-slot7cpu1# save-tech-support-info
```

Use the following optional parameters to customize the logs:

```
--nofaults --nologs --noroutinginfo --routinginfoverbose --numdays=N -
-numhours=M --datarecords --psm
```

Depending on the MCC release you are running, you may see `--nocorefiles` or `--nomemimages` instead of `--nofaults`.

For EMS issues, the *emslogdump.pl* dump file is stored in the following folder:

/opt/Affirmed/NMS/bin/./emslogdump.pl

4.1.2 Attaching Files to Tickets

The following file types are supported as attachments to a ticket:

.bmp, .css, .csv, .doc, .dwg, .eml, .gif, .gx, .htm, .html, .ico, .img, .jpg, .jpeg, .js, .mps, .mpeg, .mpg, .mpp, .pdf, .pjpeg, .png, .ppl, .ps, .qt, .rtf, .sms, .swf, .tif, .tiff, .txt, .vsd, .xls, .xml, .zip.

If the file type you want to attach is not supported:

a Add the file to a .zip file; or

b Append .txt to the filename (for example, filename.log.txt) and attach the file to the ticket.

4.2 Contacting the TAC

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:

E-mail: affirmedtac@affirmednetworks.com

24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

A Rename corefiles to fault

Affirmed Networks renamed **corefiles** to **fault**. This name change impacts filenames, traps, alarms, eventlogs, syslogs, and scripts used to capture faults and generate summary files and alarms.

> ***Note:*** *Depending on the MCC release you are running,* ***corefiles*** *is renamed to* ***fault*** *and* ***memimage*** *is renamed to* ***fault****.*

Below are examples of this renaming change:

Old Name	New Name
cluster <id> node <id> processes **core-dump**	cluster <id> node <id> processes **fault**
infrastructure oom **coredump**	infrastructure oom **faults**
infrastructure **corefiles**-filter generation	infrastructure **faults**-filter generation
infrastructure **corefile** generation	infrastructure **faults** generation
filecommand: category name for /public/ **corefiles**	filecommand: category name for /public/**faults**
file names under /var/corefiles/***core***	file names under /var/corefiles/***fault*** ***Note:*** *Existing /var/corefiles/*****core**** *files are not renamed, only new files use the new naming convention /var/corefiles/*****fault****
Alarm: anNew**Core**FileDetected	Alarm: anNew**Fault**FileDetected
save-tech-support-info script change: --**nocores**	save-tech-support-info script change: --**nofaults**
show cluster <id> node <id> **crash**-detail <**core**-filename>	show cluster <id> node <id> **fault**-detail <**fault**-filename>
show cluster <id> node <id> **crash**-summary <seconds-since-epoch> <**core**-filename>	show cluster <id> node <id> **fault**-summary <seconds-since-epoch> <**fault**-filename>

Old Name	New Name
show cluster <id> node <id> **coredump-**summary	show cluster <id> node <id> **fault**-summary
cluster <id> clear**crashes**	cluster <id> clear-**faults**
show cluster <id> node <id> **corefile**-detail	show cluster <id> node <id> **fault**-detail